Vocabulary position and trademark lifecycles:
An event-dated corpus and a lead/lag text measure for the commercial economy

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Abstract

Text-based novelty measures built on patents observe invention by the small minority of firms that patent; trademark filings time-stamp commercialization by everyone else, in plain commercial language. All 13.99 million USPTO trademark case files are re-parsed here into event-dated prosecution histories, and every filing scored on how surprising its language is against what its industry filed in the five years before it and in the five years after. The average of the two surprises is the filing's *atypicality*; their difference is its *lead*, positive for wording the industry subsequently moved toward (*leading*) and negative for wording it was already leaving behind (*lagging*).

A product described in language its industry is moving toward is less likely to still be on the market five years later than one described in language the industry is leaving behind. Among 3.10 million registrations, the fifth of marks with the most leading language fail the first use-proof declaration — the sworn statement, due between the fifth and sixth year, that the mark is still in commercial use — 53.3% of the time against 51.9% for the most lagging fifth, with class×cohort fixed effects, owner-clustered errors, and drafting and filer controls ($t = 8.3$). The penalty is episodic and technological: two to three times larger for marks filed in the late 1990s than at any time since, reversed for those filed 2000–2004, and back from 2008, with the entire swing inside four technology classes and a flat one to four points everywhere else.

0.83% of first-time filers with professional representation ever appear in SEC financial-reporting records, against 0.18% of self-filers. The atypicality of a firm's first registered mark does not predict whether its owner ever reaches SEC reporting once class, year and description length are absorbed; yet both leading ($\mu = 0.39\%$) and lagging ($\mu = 0.44\%$) owners outperform the persistent middle ($\mu = 0.24\%$). Neither axis is a proxy for patenting: within firms no association reaches a tenth of a standard deviation.

Two provisos. Atypicality can be measured on terms or on themes, and the two disagree. Term atypicality predicts both outcomes negatively (-0.09pp per σ on SEC reporting, -0.055σ on patenting); theme atypicality predicts SEC reporting not at all and patenting weakly the other way ($+0.012\sigma$). The two therefore disagree in sign on both outcomes, which makes a text measure's sign at the firm level a property of how the text was counted rather than of the firm. The second proviso is structural: every estimate conditions on the office having granted the mark, because registration is itself selected on language, and read without that conditioning the same data support weaker and sometimes opposite conclusions. Code, panels, and the corpus build are public at `https://github.com/ericsilver/tm-vocabulary`, with a browsable explorer for the per-theme and per-Nice-class results behind the pooled estimates.

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1 Introduction

Text-based measurement has reshaped how economists quantify invention. Kelly et al. (2021) score every US patent since 1840 by its textual distance from prior and similarity to subsequent patents, and Hoberg & Phillips (2016) build product-market industries from 10-K text; Kalyani (2024) measures patent creativity as the share of technical terminology not seen before and ties it to firm productivity. Both programs inherit the patent record's coverage: technical invention by the small minority of firms that patent. Most firms never patent, service-sector and business-model innovation rarely patents, and patenting propensity varies so much across industries that cross-industry comparisons are fraught (Dinlersoz et al., 2018; Graham et al., 2013). Trademark filings offer the complementary window. Nearly every commercially serious product or service introduction in the United States is accompanied by a trademark filing, the filing's goods/services description states in plain commercial language what the firm intends to sell, and the corpus extends across every sector of the economy at the rate of hundreds of thousands of granted filings per year.

The trademark–innovation literature is three decades old and has moved well past counting (Castaldi, 2020). Flikkema et al. (2025) survey its first ten years of validation work and its open questions; Willeke et al. (2026) inventory the data sources and the preprocessing burden that has kept them underused. Micro-level links to firm dynamics are established (Dinlersoz et al., 2018, 2023), and the goods/services description is already read as text rather than treated as a label: Semadeni (2006) scores consulting service-mark descriptions into a competitive positioning space, Semadeni & Anderson (2010) tracks whether one firm's described offering is imitated by rivals, and Flikkema et al. (2019) codes descriptions against the innovation each mark accompanies. Prosecution records have been exploited too: Fink et al. (2024) use contested filing sequences to study trademark races, and their outcome — continued use five years after registration — is the margin this paper's central result also runs on. What Flikkema et al. (2025) put as the field's next step is the move from counts and innovation intensity to the *type*, *quality*, and *trend* content of what marks cover: the trademark analogue of what Kelly et al. (2021) and Kalyani (2024) did for patent text. That analogue does not yet exist at corpus scale, and the two strands above — description text and prosecution events — have run on separate data.

This paper builds both on one corpus, so that what a filing says and what subsequently happened to it are observed on the same record.

Whether a description's language is strange for its industry predicts very little; whether it arrived *before* its industry's language moved that way predicts a good deal. Marks whose wording the industry subsequently adopted are abandoned at the five-year use-proof gate more often than marks written in language the industry was already leaving behind. Their owners also reach SEC financial reporting less often than the owners of the most lagging marks, though both tails do better than the middle.

The USPTO distributes its trademark backfile as bulk XML — 83 archives covering 1884 to 2025 — from which the published research files retain a filing's current status but not the dated sequence of events that produced it. Re-parsing the backfile yields 242 million prosecution events — every step in the examination of an application — across 13.99 million case files, each event carrying a code and a date: every office action, response, maintenance filing, and cancellation. The event codes are resolved into a 760-code dictionary. A header pass extracts counsel of record, the statutory filing basis, and the postregistration declarations. A third pass recovers owner domicile, which the class-level files omit. Owner names are then normalized to a clustering key and resolved to two external universes: SEC registrants, through the agency's own company-name and ticker files, and Regulation D issuers, through the Form D quarterly data sets — the latter filtered to operating companies, with ambiguous names dropped rather than guessed. That link is what lets a mark's language be followed to financing and to public listing. Where the patent record dates invention once, at grant, the trademark prosecution record dates a commercial claim's entire life, including the use-proof gates at which marks attached to products that never materialized are culled.

Every filing is grouped by its industry class — one of the 45 international goods and services categories — and placed by its filing date, then put to two readers who have seen disjoint halves of that industry's stream: one who knows only what the class filed before it, one who knows only what it filed after. Each reader's "how surprising" is a Kullback–Leibler divergence between the filing's mix of topics — its weights over word-clusters fitted to the corpus — and the industry's, large when the filing puts weight where its industry's filings put little. *Atypicality* is the average of the two answers. *Lead*, written ΔKL, is the gap. Language that surprised the first reader and not the second was surprising because it came first: the industry moved toward it. Language that read as ordinary to the first and unusual to the second was left behind as the field moved on. Filings of the first kind are *leading*, of the second *lagging*.

The construct adapts Murdock et al. (2017) and Barron et al. (2018); Section 3 sets out what transfers and what does not. The nearest measurement precedent on this corpus is von Graevenitz et al. (2022), who flag goods/services tokens that are both new and fast-spreading; their unit is the token and their flag binary, while the score here is continuous and attaches to the filing, so every application carries a position that can be crossed with its own prosecution outcomes.

Results are stated on marks the office granted. Registration is itself selected on language, so a correlation over all filings mixes what examination does to a description with what the market does to the product; Appendix E reports the unconditional version. Two further products of the exercise are released with it: the corpus, and a measurement hazard that applies to text-based novelty measures generally.

2 Registration measures follow-through; the use-proof gate measures survival

A US trademark passes through a sequence of dated, high-stakes steps, and each step leaves a record. An application is filed with a goods/services description; an examiner reviews it, often issuing office actions the applicant must answer; if allowed and (for intent-to-use filings) a statement of use is filed, the mark registers. Registration starts a maintenance clock: in years five to six the owner must file a Section 8 declaration proving continued use in commerce, with a six-month grace period; around year ten a combined Section 8 declaration and Section 9 renewal is due, and every ten years thereafter. A mark that misses a use-proof gate is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout. Graham et al. (2013) describe the administrative data; the event-dated corpus built here adds the full prosecution history of each case, so a cancellation is assigned to a specific gate by its date rather than inferred from a status snapshot.

Scoring runs 1995–2019, because a full five-year past reference does not exist before 1990 and a full forward reference does not exist after 2020; the analysis cut of 1995 is about two years more conservative than the measured burn-in requires (Appendix A). The gate cohorts end in 2018 because the cancellation record runs to April 2026 and the failure window closes at registration age 8.5 years, so 2018 is the last cohort whose window has essentially elapsed — the top of that range is fractionally censored, and a separate 2019 replication is reported below. They begin in 2002 because that is the conventional cut for this corpus, and that lower bound turns out to matter: carried back to the first scoreable cohorts the penalty is two to three times larger than it has ever

Table 1: The trademark funnel in this corpus. Registration counts are unique serials; gate outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Sources: the TRTYRAP re-parse and the scored filing panels described in Section 3.

Stage	n	Share
Debut filings scored, 1995–2018 (owners' first filings)	3,589,068	
reached registration	2,066,600	57.6%
All unregistered class-records filed 1995–2018	3,798,318	
never filed statement of use		42.6%
stopped responding in prosecution		45.3%
removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,534	
failed the first use-proof gate (event-dated)		52.3%
Passed first gate, cohorts 2002–2013, terminal status known	1,129,724	
failed at the year-ten renewal		39.1%

been since, and Section 4.3 reports what the restricted window concealed.

Registration is a launch indicator rather than an examiner verdict: among applications that fail to register, roughly nine in ten die by the applicant's own inaction — never filing the statement of use, or going silent during prosecution — and only 4.3% are removed adversarially (Table 1). The Section 8 gate is where the market's verdict reaches the record: half of registered marks do not survive it, and failing it means the product the mark named is no longer in commerce.

The registration step is analyzed as its own outcome; the gate results condition on it; the unconditional composite, which stamps the registration step's shape onto any downstream outcome, is confined to Appendix E. Outcomes are read from the event-dated prosecution history, which attributes each cancellation to a specific gate and observes counsel and basis directly, and cross-checked against the USPTO status-code dictionary.[1]

3 The construct: how surprising a description is, and when

3.1 Two readers, one living backwards

Imagine two readers of a filing's goods/services description, each having seen a disjoint half of its industry. One has read only what the industry filed in the five years before it. The other, living backwards as Merlin was said to, has read only the five years after. Each is asked how surprising the wording is. The measurement is how much they disagree: a description that startles the first reader and not the second arrived before its industry's language moved that way.

Both windows are anchored on the filing's own date — the 1,826 days before it and the 1,826 days after — so no two filings made on different days share a reference, and filings made on the same day fall in neither of the other's windows. Inside a window every day counts the same, so a term the class adopted five years ago and one it adopted last month are equally familiar. The measure asks whether wording is unusual against a period, not whether it is ahead of a trend inside that period; it has no momentum channel.

One reference per class-year, the alternative, scores a January and a December filing against the same object and lets neither see any of the language filed during its own year. That imprints a

[1] 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired.

spurious gradient: mean lead then rises almost monotonically with filing month, purely because a December filing sits eleven months further from its past reference and nearer its future one. It cost roughly a third of the measured gate penalty and the whole of an apparent curvature at registration (Appendix A.5).

3.2 From a description to two numbers

The apparatus is that of Murdock et al. (2017), developed on Darwin's reading notebooks, and Barron et al. (2018), who extended it to French Revolution parliamentary debates. They score a document against those before it and again against those after, calling the two quantities *novelty* and *transience* and their signed difference *resonance*. The arithmetic carries over unchanged. The interpretation does not: their document has one author and the reference stream is that author's own reading, so a high value says something about a mind, while here a filing is scored against other firms' filings and its author is usually counsel drafting for scope of protection.

Counting words. A description is reduced to a bag of words: the text is lowercased, split into tokens of three or more letters, and counted, with adjacent pairs counted as tokens in their own right so that *computer software* registers alongside *computer* and *software*. Order is discarded. Take the filing for AMAZON.COM, lodged in the advertising and retail class on 18 April 1997:

> computerized on line search and ordering service featuring the wholesale and retail distribution of books, music, motion pictures, multimedia products and computer software in the form of printed books, audiocassettes, videocassettes, compact disks, floppy disks, CD ROMs, and direct digital transmission

Of its 45 distinct terms, 33 are in the theme model's vocabulary, among them *computerized*, *search*, *ordering*, *ordering service*, *retail*, *distribution*, *printed books* and *digital transmission*.

Grouping words into themes. Scoring those tokens directly is possible, and the paper reports it as a cross-check, but on documents this short a raw term count is dominated by length: the fewer words a description has, the spikier its distribution and the further it sits from any reference, so "unusual" and "brief" become nearly the same measurement — term-scored atypicality correlates with log length at −0.729 in the software class, against −0.251 under themes. The primary scoring therefore groups the vocabulary into 50 themes fitted across the corpus by latent Dirichlet allocation and represents a filing by how its tokens distribute over them. The Amazon filing puts 0.40 of its weight on a theme led by *video, computer, electronic, music, games, audio*, 0.15 on one led by *services, computer, software, information, data*, and the rest across the remaining 48.

Themes are what make combinations visible, and also what limit them. A filing built entirely of ordinary words can still land where its industry is not, because the *mixture* is unusual — the Amazon case, where every token is unremarkable and the invention lives in *on line search and ordering* applied to books. But a theme basis fitted on the corpus has no coordinate for language the corpus has not yet seen, so a genuinely new word arrives invisible: KOMBUCHA in 1997 is a drink with no theme of its own and reads as an ordinary beverage until enough filings mention it to move the fit. Term scoring sees the new word and misses the new mixture; theme scoring does the reverse. Appendix G works three such filings through both.

Comparing against the two references. Write P for the filing's distribution over themes, and Q_{past} and Q_{future} for that distribution averaged over every same-class filing in the two windows,

$[d_i - W,\, d_i)$ and $(d_i,\, d_i + W]$, with $W = 1{,}826$ days. How far P sits from a reference Q is the Kullback–Leibler divergence,

$$\mathrm{KL}(P \,\|\, Q) \;=\; \sum_k P_k \log \frac{P_k}{Q_k},$$

a sum over the 50 themes, zero when the two distributions agree and growing as the filing puts weight where the industry puts little. It is measured in nats, the natural-log unit. Write

$$K^- = \mathrm{KL}(P \,\|\, Q_{\text{past}}), \qquad K^+ = \mathrm{KL}(P \,\|\, Q_{\text{future}}),$$

so K^- is what the first reader feels and K^+ the second. The source literature calls these *prospective* and *retrospective* KL, naming the reader's vantage rather than the window's direction.

Taking the difference. Amazon's filing is scored against 41,166 earlier filings and 146,532 later ones, giving $K^- = 1.286$ and $K^+ = 1.163$. The two axes used throughout are their average and their signed difference:

$$\underbrace{A \;=\; \tfrac{1}{2}\left(K^- + K^+\right)}_{\text{atypicality}}, \qquad \underbrace{L \;=\; K^- - K^+}_{\text{lead}}.$$

For Amazon, $A = 1.225$ and $L = +0.123$. Against its own class those say something the raw pair does not: its atypicality is at the 47th percentile, entirely ordinary, while its lead is at the 94th. The industry moved toward this description in the five years after it was filed and had not been writing that way in the five years before. Nothing about the wording was strange; it was early. L is signed, positive for a leading filing and negative for a lagging one, and the symbol $\Delta\mathrm{KL}$ is used interchangeably with it.

Differencing is what makes that separation possible. K^- and K^+ correlate at 0.988 on the software class, so entering both is close to entering one variable twice; rotating them into an average and a difference isolates the large common component from the small residual that carries the timing, and leaves the two regressors nearly uncorrelated (+0.036 against each other, where K^- alone would give +0.114). The cost is that L is a small residual of two large quantities, about a sixth of either level's spread, which Appendix A quantifies as a genuine fragility.

Table 2: The quantities, their sources, and the terms used here. *Lagging* and *leading* describe the two signs of L; *atypicality* is unsigned and says nothing about direction.

Quantity	Definition	Murdock et al.	Term used here
Surprise against the past	K^-	novelty	past-facing surprise
Surprise against the future	K^+	transience	future-facing surprise
Their average	$A = \frac{1}{2}(K^- + K^+)$	—	**atypicality**
Their signed difference	$L = K^- - K^+$	resonance	**lead** (leading / lagging)
Its magnitude	$\lvert L \rvert$	—	lead magnitude

3.3 Within class, not across it

Goods/services descriptions are drafted to secure scope of protection, not to describe a product distinctively, and counsel reuse language — from the USPTO's own Acceptable Identification of Goods and Services Manual, from house precedent, and from competitors' successful filings. Identical wording across unrelated firms is therefore common, and it carries through to the scores: 17% of scored registrations share a value with another filing in the same class and cohort, which is why the tie rule is stated and tested (Section 4.2). Within a class, the measure picks up departure from a shared drafting convention. Across classes it does not: a class where the Manual supplies dense standard language shows a compressed distribution of atypicality, and one where firms describe novel services in prose a wide one, for reasons unrelated to how innovative either industry is. Every estimate is therefore taken within class and year, and magnitudes compare across classes only as signs and orderings.

The vocabulary is positional rather than causal. A leading filing was ahead of its industry's language, which implies a trajectory without asserting the filing caused it. *Atypicality* is likewise textual, and is not trademark law's distinctiveness doctrine, which governs registrability along the generic-to-fanciful spectrum; the two attach to different objects, one to the mark and one to the description of the goods it covers.

4 Among granted marks, being early costs and being unusual does not pay

4.1 Why these results condition on grant

Registration is not a neutral filter on language, so every estimate below conditions on it. Completion is an inverse-U in atypicality: applicants whose language sits far from their category in either direction follow through least often, and the penalty is larger for the conventional tail than the unusual one. Pooling granted and abandoned filings therefore mixes what examination does to a description with what the market does to the product, and the resulting gradients are attenuated and in places reversed (Appendix E).

Conditioning would be the wrong move if abandonment were usually a pause rather than an ending — if firms reworked refused applications and refiled them, the same commercial intent would be counted once as a failure and again as a success. It is not. Among 2.28 million abandoned applications, only 3.6% are followed within six years by the same owner filing the same mark again in the same class, and only 1.5% by one that then registers. Crediting every successful refiling as an eventual registration moves the inverse-U in atypicality from a depth of +2.91pp to +2.85pp. Abandonment is overwhelmingly terminal, and what the registration step selects on is therefore a property of the filing rather than an artifact of counting.

What examination rewards is a middling specificity — concrete enough to survive a definiteness objection, conventional enough to avoid a likelihood-of-confusion refusal. That is a fact about examiners, and Appendix D sets it out in full: the shape on each axis, the compositional trough on the signed axis, and why familiarity alone cannot produce it. The rest of this section is about what happens to marks after the office has granted them.

4.2 Leading marks fail the use-proof gate

Marks written in language their industry subsequently moved toward fail their first use-proof gate more often than marks written in language it was moving away from. Within class×cohort cells, with owner-clustered errors and drafting and filer controls, marks in the top ΔKL quintile fail +1.4pp more often than those in the bottom ($t = 8.3$): percentage points on a 52.3% base rate, not percent of it, so 53.3% failure in the most leading fifth against 51.9% in the most lagging.

About seventy percent of it sits in the top quintile, where the step from the fourth is seven times the average of the three below it (Figure 2, panel A). This is a threshold rather than a gradient, so estimates below are top-versus-bottom contrasts throughout.

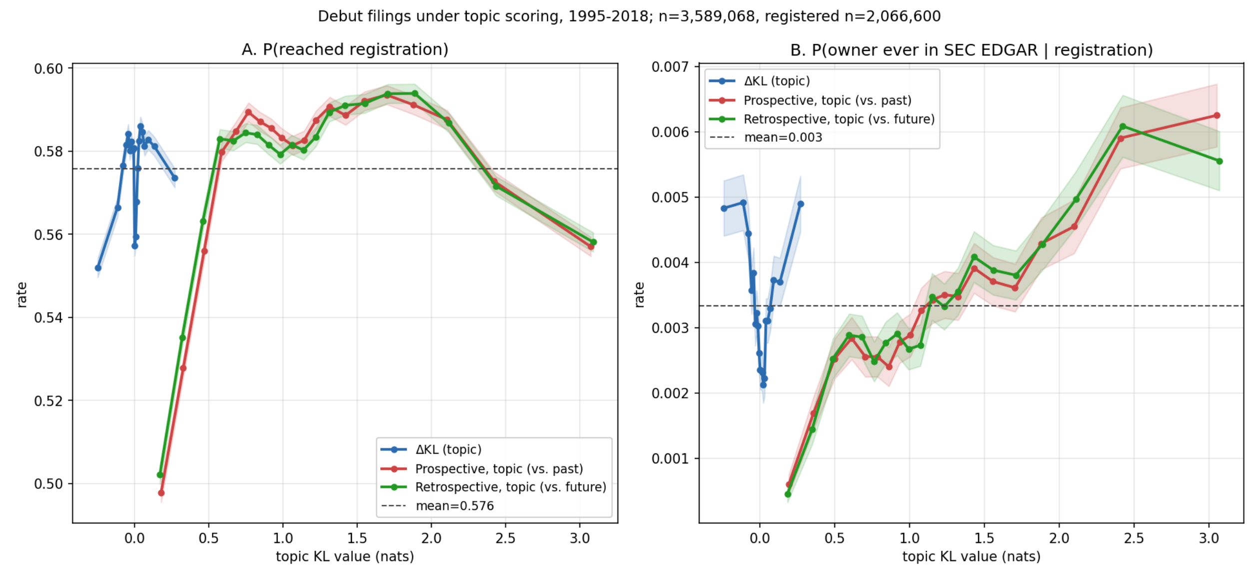

Figure 1: Outcomes for debut filings (owner's first-ever filing) across all 45 Nice classes under topic scoring, filing years 1995–2018, in quantile bins with 95% intervals. $n = 3{,}589{,}068$; registered subset $n = 2{,}066{,}600$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR | registration); the structure in panel B is decomposed in Section 4.4.

Estimates are on gate *failure* unless the text says survival. The gate is the first maintenance filing: Section 8 for domestic registrations and Section 71 for Madrid Protocol §66(a) extensions of protection, both due between the fifth and sixth anniversary. Both cancellation codes are read. Scoring only the Section 8 code records every §66(a) registration as a survivor whatever became of it, which puts a spurious −46pp coefficient on them; reading both moves it to +10pp, since Madrid registrations in fact fail this gate more often than domestic ones.

Among 4.12 million registration class-records 2002–2018, dated first-gate cancellation rises from 50.0% at the bottom topic-ΔKL quintile to 52.4% at the top, positive in 33 of 44 classes with sufficient volume (Figure 3). But a raw pooled contrast mixes the within-industry gradient with class and cohort composition, treats multi-class registrations as independent observations, and ignores that gate outcomes correlate within owner. The primary estimates therefore come from a linear probability model on 3.10 million unique registrations, one row per serial, with ΔKL quintiles cut *within* class×cohort cells, a cohort here being a year of registration. The specification adds class×cohort fixed effects; controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile, and foreign filing basis;[2] and standard errors clustered on 1.27 million normalized owners (Table 3).[3]

[2] A US application is filed either on *use* — the mark is already in commerce — or on *intent to use*, which defers proof until after allowance. Foreign applicants may instead file on a home registration (§44(e)) or extend an international registration into the US under the Madrid Protocol (§66(a)); the latter maintain under §71 rather than §8. These are statutory routes to the same registration, and they differ sharply in who files and how often the mark is abandoned, which is why each enters as a control.

[3] Ties are common. Within the class×cohort cell the quintiles are cut in, 17.1% of scored registrations share a ΔKL value with at least one other filing, and the largest tie group runs to 53. Anchoring the reference window on the filing date does not remove them, because filings that share a class, a filing date and a goods/services description — an owner registering a family of marks at once, on Manual language — have identical distributions scored against identical references. Cut points are fixed by sorting on the score and then on the serial number, which is reproducible but systematic: serials run in filing order. It makes no difference. Sorting the ties the opposite way moves the top-

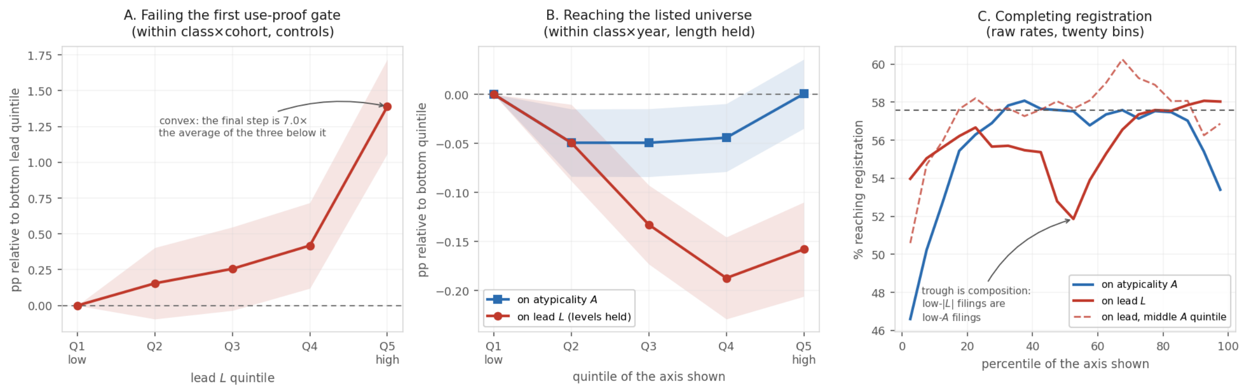

Figure 2: Quintile profiles for the three outcomes, on a common axis. Shaded bands are 95% intervals. *A:* first-gate failure against lead, within class×cohort with controls, relative to the omitted bottom quintile — convex, with the penalty concentrated in the top quintile. *B:* becoming SEC-reporting, within class×year with length held — flat in atypicality once length is absorbed, and falling in lead through the fourth quintile once the levels are controlled. *C:* raw registration rates for debut filings in twenty bins — an inverse-U on atypicality, and on lead a trough just above the median that is compositional, absent once atypicality is held in its middle fifth (dashed). Horizontal line at the pooled rate.

The gradient is monotone within cells. The full control set cuts the magnitude by about a third without absorbing it, and the estimate is if anything *largest* in the cleanest stratum — counsel-represented, US-domiciled owners — so it is not carried by the self-filed or foreign mass-filing segments whose gate failure rates are high for their own reasons.[4] About forty percent of the raw pooled contrast is composition: +1.4pp design-based against +2.5pp raw.

Within the 2002–2018 window the penalty looks modern: indistinguishable from zero for the 2002–2005 cohorts, turning positive in 2006, and running +1.8 to +5.6pp for every cohort from 2008 onward (Figure 4). That reading does not survive carrying the series back, and Section 4.3 replaces it.

The gate is close to a fixed appointment rather than a hazard spread over years: of the 1.35 million cancellations recorded for fully observed cohorts, 17 post before registration age 6.5, 96.1% post between 6.5 and 7.0, and the rest after seven. Since the cancellation record ends on 2 April 2026, a cohort is observable only once it has reached seven years, which makes 2019 the last one with usable gate data and leaves 2020 and later with no observed gate at all. Measured on the 6.5–7.0 band, fully observed for every cohort it covers, the penalty runs between +2.1 and +5.6pp with no trend across the decade and overlapping intervals at its two highest points. The 2019 reading rests on the 78,408 registrations issued early enough in that year to reach seven years inside the record, so it is a first-quarter subsample rather than a cohort.

Because the outcome is an indicator on a window, the regressor could in principle be predicting when marks fail rather than whether they do, and a fixed window would record the first as though it were the second. It is not. On the 3.15 million registrations whose ninth anniversary is inside the record — for which failure is settled and no window is needed — the contrast is +2.71pp within

versus-bottom contrast by 0.001pp, and five pseudorandom orderings move it by at most 0.002pp, against a contrast of 2.287pp — a tenth of one percent of the estimate.

[4]The level effects on those covariates are large in their own right: counsel −19.5pp and Madrid §66(a) basis +10.3pp.

Table 3: First-gate failure and ΔKL: design-based estimates. LPM of event-dated first-gate failure on within-class×cohort ΔKL quintiles (Q1 omitted), unique serials, registrations 2002–2018. Failure is a dated cancellation for non-use at registration age 4.0–8.5 years, under §8 or, for Madrid §66(a) registrations, §71. All specifications include class×cohort fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). Base failure rate 52.3% on this sample.

Sample	Specification	Q5−Q1 (pp)	SE	n
All	FE only	+2.29	0.16	3,104,534
All	FE + controls	+1.39	0.17	3,104,534
Counsel-represented	FE + controls	+1.15	0.20	2,414,318
Counsel + US-domiciled	FE + controls	+1.60	0.22	1,888,352
Self-filed	FE + controls	+1.80	0.26	690,216
Excluding Madrid §66(a)	FE + controls	+1.82	0.18	2,924,427
Counsel + US, 2002–07	FE + controls	−0.25	0.33	551,989
Counsel + US, 2008–12	FE + controls	+2.00	0.42	551,924
Counsel + US, 2013–18	FE + controls	+2.65	0.25	784,439
All	continuous, pp per σ	+0.58	0.05	3,104,534
Counsel + US	continuous, pp per σ	+0.63	0.06	1,888,352

Standard errors replace significance stars: with n in the millions every coefficient has $p < 0.01$, so stars would mark each row identically. Errors are clustered on *normalized owner* — names uppercased, stripped of punctuation and of twenty legal suffixes, so ACME CORP. and ACME CORPORATION are one cluster. A single owner files many marks, drafts them in a house style, and abandons them together when the business fails; the within-owner correlation of gate survival is 0.42, so the 1.27 million owner clusters, not the filings, carry the information. The Madrid row drops §66(a) registrations rather than absorbing them in a control, since they maintain under a different statute; the estimate is larger without them.

class and cohort, against +2.65pp on the paper's 4.0–8.5 window and +2.65pp on the narrow band. Conditional on failing, the probability of failing late carries a contrast of −0.01pp ($t = -0.2$) once class and cohort are absorbed. The pooled version of that contrast is +0.27pp and is composition.

The filer population changed over the same period. Within counsel-represented US-domiciled owners, and with the full control set, the earliest cohort third is slightly negative (−0.25pp, SE 0.33) while the two later thirds are clearly positive (+2.00 and +2.65pp, Table 3). The gradient is therefore not an artefact of the post-2015 surge in self-filed and foreign applications.

4.3 The penalty is not modern, and it is not general

Scores exist from 1995 on, which makes registration cohorts from 1996 evaluable, and their gate outcomes have been settled for two decades. Carried back, the contrast runs +6.9pp for the 1996 cohort and rises to +10.1pp for 1999 — two to three times anything observed since — then collapses. The 2002–2005 cohorts the paper had been reading as a quiet baseline are a trough between two episodes, and one of them is the larger.

Scores begin in 1995, so a 1996-cohort registration must have issued within a year of filing; early cohorts are selected on prosecution speed, and speed plausibly travels with drafting in standard language. Holding pendency in a common one-to-three-year band leaves the series unchanged. Indexing by filing year instead, so a cohort contains every prosecution speed, gives +7.9 to +9.6pp for filings 1995–1999 and −0.95pp for filings made in 2000 — a ten-point swing in a single filing year.

Splitting the four classes that carry most technology filings — 009, 035, 038 and 042 — against the other forty-one separates a large moving component from a small steady one (Figure 5):

Filing years	Technology classes	All others
1995–1999	+14.18 (0.36)	+4.48 (0.27)
2000–2004	−1.37 (0.35)	+1.21 (0.24)
2005–2007	+2.04 (0.43)	+1.99 (0.28)
2008–2014	+9.02 (0.25)	+1.32 (0.17)

Outside technology the penalty is a flat +1 to +4pp in every era, including the one in which the pooled series reads as zero. Inside technology it swings by fifteen points: strongly positive for filings made into the late 1990s, negative for those made in the five years after, and positive again from 2008. The pooled null of 2002–2005 is a technology reversal partly offsetting a non-technology positive, not an absence.

What this design cannot say is which cycle the swing belongs to. Marks filed 1995–1999 reach their gate in 2001–2007 and marks filed 2000–2004 reach theirs in 2006–2013, so a story in which unusual language is punished because it was written during an investment boom and a story in which it is punished because the gate falls during a bust both fit these numbers.

Levels and gradients separate cleanly across filer strata. Self-filers fail the gate far more often than represented owners (69.6% versus 46.8%), and intent-to-use registrations more often than use-based ones (53.1% versus 50.2%), but the ΔKL gradient itself is similar across these strata once class, cohort, and drafting covariates are held fixed (+1.8pp self versus +1.2pp counseled in Table 3). Per class, the largest penalties sit in tobacco (+26pp, the vaping era), hand tools (+10pp), processed foods (+10pp) and medical apparatus (+9pp), with negative lifts in beverages, transport and construction.

The risk persists at the year-ten renewal, and on the design-based estimates does not attenuate: the lead coefficient is −0.49pp at the first gate and −0.63pp at the second (Table 5). What attenuates is the raw single-class contrast. In the software diagnostic class, where the event-dated first-gate lift on the 2010–2013 cohorts reaches +12.1pp, the conditional lift at the second gate falls to +7.6pp; pooled across classes the second-gate profile is a shallow U. By year ten the lagging tail has closed the gap in cumulative failure, having simply taken longer to get there. Most of the difference resolves at first proof of use.

A competing reading is that the gate measures disengagement rather than products: an owner who has lost interest both answers the examiner slowly and lets the registration lapse. The prosecution record tests this through the gap between the first non-final office action and the response to it.

Response speed does not predict the gate. Across the 1.22 million scored registrations with a usable office-action/response pair, gate passage varies between 46.1% and 48.1% across response-speed quintiles, from a median of seven days to 183 — flat, and if anything the quickest responders pass slightly *less*. The lead penalty is present in both halves: −3.48pp (±0.40) among faster-than-median responders and −2.36pp (±0.40) among slower, against −2.94pp pooled. The restriction drops 61% of the sample, disproportionately applications that never drew a refusal, so this speaks to the prosecuted subset.

Mixing the past and future reference windows identifies which kind of novelty the gate punishes. The penalty loads on breaking with the established past (−7.3pp) rather than on resembling the future (−3.2pp), and in software the separation is complete (−12.7pp against −0.3pp). This decomposition runs on annual reference buckets, since it requires a separate scoring at each of $W \in \{3, 5, 7\}$ on both sides; by Appendix A.5 its magnitudes are likely inflated by about a third, so only the contrast between configurations is relied on (Appendix A).

4.4 Being early costs at the public-reporting margin; being unusual does not pay there

The outcome here is whether an owner ever appears in the Securities and Exchange Commission's financial reporting records. That is broader than exchange-listed: roughly half the matched firms carry a ticker and the rest report for other reasons. The rate rises monotonically in the KL *levels* on both axes, from 0.19% at the bottom past-facing quintile to 0.52% at the top. In signed ΔKL it is U-shaped.

Table 4: Debut listing and vocabulary position: design-based estimates. LPM of P(owner ever in SEC EDGAR) on 1,556,968 registered debut filings 1995–2018, one observation per owner, class×filing-year fixed effects, heteroskedasticity-robust SEs. Quintiles cut within class×year. Base rate 0.481%.

	Coefficient (pp)	SE
Atypicality quintiles, $A = \frac{1}{2}(K^- + K^+)$ *(Q1 omitted)*		
Q5, FE only	+0.017	0.018
Q5, FE + log length	+0.000	0.018
Q5, FE + log length, on K^- alone	−0.003	0.018
Signed ΔKL quintiles, FE + log length + KL levels		
Q3	−0.133	0.020
Q4	−0.187	0.021
Q5	−0.158	0.024
Decomposition, FE + log length		
|ΔKL| (per σ)	+0.067	0.009
signed ΔKL (per σ)	−0.047	0.006
log description length	+0.056	0.004

Standard errors in place of significance stars; see the note to Table 3.

The raw atypicality gradient does not survive the design-based estimates (Table 4). Within class and filing year, holding description length fixed, top-quintile atypicality raises the probability of SEC reporting by +0.000pp on a 0.481% base ($t = 0.0$), and by −0.003pp on the past level alone. The quintile profile is not even monotone: the three middle quintiles sit *below* the bottom one and the top merely returns to it. The 2.7-fold gradient in the raw bins is composition — firms that go on to report cluster in particular classes, years, and multi-class filings — and class×year fixed effects absorb it. Lead costs, at equal atypicality: the signed component enters negatively (−0.047pp per σ, $t = -7.8$) while the unsigned component enters positively (+0.067pp per σ, $t = 7.7$). The U-shape in raw bins is exactly this pair: lagging filings show the atypicality premium with no lead penalty, leading filings show it net of that penalty, and the middle quintiles show neither. Once the levels are controlled, the apparent recovery of the far-leading tail is much reduced: the quintile profile falls through Q4 and recovers only slightly at Q5 (Table 4 middle panel).

Owners are matched to SEC registrants by normalized exact name, over a candidate universe of all 5.67 million owners in the corpus; names resolving to more than one registrant are dropped rather than assigned, and one match is kept per owner string. That yields 19,889 matched owners covering 7,588 registrants, and a base rate of 0.481% among registered debut filers. Validation on 22 companies that indisputably went public — across software, pharmaceuticals, food, apparel, aerospace and autos — matches all 22. A random sample of 25 matches carrying at least three filings was inspected by hand and none was wrong, which is a check on gross failure modes rather than a precision estimate.

The linkage is nonetheless a name match, and that bounds the atypicality reading in particular. Firms with distinctive names are easier to match than firms with generic ones, and distinctiveness of a name plausibly travels with distinctiveness of the goods/services text; nothing in this design separates the two. Restricting the candidate universe — to a subset of classes, say — makes matchability itself a function of the regressor, and the atypicality gradient is sensitive to exactly that. Its absence here is therefore the more credible reading of the two, and the signed axis, which does not move when the linkage is varied, carries the weight. The financing rows of Table 5 rest on a Form D match whose raw extracts are not in the current tree and should be read as provisional.

In raw rates, 0.444% of the most lagging fifth of registered debut owners later appear in SEC reporting records, against 0.240% in the middle fifth and 0.390% of the most leading fifth: both tails beat the middle, and the lagging tail beats the leading one. Under the design-based specification the profile declines from the lagging end through the fourth quintile and recovers slightly at the leading end, so the ordering is monotone over four of the five quintiles.

The profile is a U rather than a slope, so a linear summary of it points downward through a curve whose tails are its high points; Appendix B fits the shape. And "SEC reporting" is not "went public": splitting matched owners on whether their CIK carries a ticker gives 10,556 listed against 9,333 reporting without one, and both halves show the same U, so the result is not an artifact of counting non-operating registrants as successes.

Atypicality predicts survival at the use-proof gate but not which owners become SEC-reporting once composition is absorbed. Lead is the timing bet, and it costs at both margins.

The two are not independent in the obvious economic sense. What makes a filing leading is precisely that its industry moved toward its language, and an industry that has moved toward your language now describes its goods the way you describe yours. Being early therefore consumes the quantity that pays: the mark that was unusual in 2011 is ordinary by 2016 not because its owner changed anything but because the category arrived. On this reading atypicality is a position competitors erode, and lead measures how fast. The design cannot demonstrate that mechanism — it cannot separate a category catching up from a common cause moving both — but it is what a negative coefficient on lead at equal atypicality would look like if the mechanism were real.

4.5 The construct is not patenting

Within the firms that do both, trademark lead and patenting move independently. On a panel of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024), a fixed-effects regression of standardized firm-year mean ΔKL on log(1 + patents), with firms absorbed and errors clustered on the firm, gives -0.058σ (SE 0.006, $t = -9.4$) on 134,075 matched firm-years across 17,506 firms — and that estimate does not survive a change of representation:

Representation	Reference weighting	β (σ)	t
Topics	flat window	−0.058	−9.4
Topics	two-year half-life	−0.057	−9.3
Terms	flat window	+0.012	+2.0
Terms	two-year half-life	+0.009	+1.5

The weighting of the reference is irrelevant — flat and decayed agree to a thousandth — but the representation flips the sign. No estimate anywhere in this comparison reaches a tenth of a standard deviation, so a firm moving from no patents to twenty shifts its mean lead by at most a fifth of one, and the direction of even that depends on whether one scores topics or terms.

One objection is timing: a patent is applied for before its invention is in public use and a trademark when the firm is ready to sell, so patents would lead. Firms do not proceed that linearly, and in some industries the trademark comes first (Seip et al., 2018). Entering patents at $t-2$, $t-1$, t and $t+1$ — the last allowing the trademark to precede the patent — one offset at a time, since the count is strongly autocorrelated within firm, gives -0.054 to -0.089σ under topic scoring and $+0.012$ to $+0.024$ under term scoring. No ordering isolates a direction: the offsets are flat within each representation and opposite in sign between them. The largest coefficient anywhere is under a tenth of a standard deviation.

The second objection is heterogeneity: a pooled near-zero is consistent with genuine orthogonality everywhere, but equally with a positive slope in sectors where the two rights are complements — pharmaceuticals, chemicals, medical devices, machinery — offsetting a zero or negative slope where trademarks do the work alone. Splitting the panel on a complement/neutral/substitute grouping fixed in advance — assigned by judgement from the class definitions rather than estimated, with the eleven classes covering chemicals, pharmaceuticals, machinery, electronics, medical apparatus and scientific services treated as complements, the consumer-goods and services classes as substitutes, and the remainder neutral — and re-estimating the identical specification inside each cell, gives -0.000 ($t = -0.03$) among complements, -0.026 ($t = -1.45$) among substitutes, and an interaction of -0.008 ($t = -0.49$). Across the 27 Nice classes clearing a 50-owner floor, the complement classes run from sixth place to twenty-fourth by coefficient and do not agree in sign with one another. Within that 27-class set the only cells that reach significance are paper and housewares, both negative and neither a complement class. On the wider 34-class retention used for the sectoral build, the significant positives are transport and cosmetics — again neither a complement — while medical devices and machinery turn up among the significant negatives. Neither retention supports the complementarity story.

The test is conditional on a successful assignee match, so every owner patents at least once and the complement and substitute cells resemble each other more than the industries they stand for do, pushing any interaction toward zero. The weight rests on the cell-level nulls — among the firms where a patent–vocabulary relationship should be easiest to find.

4.6 Each axis bites at a different stage

Table 5 re-estimates every outcome on one harmonized sample — one row per debut owner, scored on that owner's first filing — under a single specification, so the coefficients can be read down a column. Magnitudes differ from the section-specific tables above.

Surprising language costs at registration and pays afterwards: atypical debut filings complete registration less often (-1.35pp per standard deviation), then survive the first use-proof gate more often ($+1.35$pp) and reach public markets more often ($+0.04$pp on a 0.56% base — the one specification in which a positive atypicality coefficient on listing survives, and it is a continuous term through a non-monotone profile, not the quintile contrast of Table 4). The examination process penalizes surprise; the market does not. Having been early runs the other way: leading applications complete registration *more* often ($+0.87$pp) and then pass the use-proof gate *less* often (-0.49pp on passing). Both survival rows point the same way at the year-ten renewal, four years later and on a sixth of the sample: atypicality $+1.30$pp and lead -0.63pp among marks that had already cleared the first gate. Whatever the first gate is selecting on, the second gate selects on it again rather than reversing it. Those filers follow through on the paperwork; what does not follow through is the product.

A linear coefficient cannot represent the inverse-U documented in Section 4, where both vocabulary tails complete registration less often than the middle. The positive linear term reflects the asymmetry of that shape — the leading tail completes more often than the lagging tail — and not a monotone gain from leading.

Table 5: Every outcome in the funnel, conditioned on its prerequisite, under one specification. Unit is the debut owner, scored on that owner's first filing. Each row is a linear probability model with class×debut-year fixed effects and heteroskedasticity-robust errors; coefficients are percentage points per standard deviation. *Atypicality* is the average of the two KL levels; *lead* is the signed difference. Cohort windows differ because each maintenance gate needs elapsed time and Form D is electronic only from 2009, so the rows are not a nested decomposition. The first gate is a dated cancellation for non-use; the second is whether a registration that cleared the first was renewed rather than cancelled or expired, restricted to 2002–2013 cohorts, since a year-six and a year-ten death share a terminal status and are separable only this way. This table alone is estimated at $T = 200$ topics rather than the $T = 50$ used elsewhere; the sign and ordering of every row are unchanged at $T = 50$.

			Unsigned		Signed
Outcome *(population)*	n	Base	Atyp.	$\lvert\Delta\mathrm{KL}\rvert$	Lead
Reached registration *(filed)*	3,161,552	56.11%	−1.352 (0.033)	−0.307 (0.044)	+0.872 (0.028)
Passed gate 1, yr 6 *(registered)*	1,260,392	41.82%	+1.349 (0.054)	−1.150 (0.082)	−0.487 (0.052)
Passed gate 2, yr 10 *(gate 1)*	297,944	55.92%	+1.296 (0.108)	−0.297 (0.162)	−0.629 (0.105)
Raised Reg D round *(filed)*	1,535,004	2.53%	+0.003 (0.015)	+0.148 (0.027)	−0.149 (0.016)
Raised Reg D round *(registered)*	916,448	2.61%	−0.017 (0.020)	+0.115 (0.035)	−0.114 (0.021)
Listed after round *(funded)*	38,886	3.49%	+0.493 (0.129)	+0.608 (0.170)	−0.513 (0.104)
Became SEC-reporting *(registered)*	1,773,938	0.56%	+0.042 (0.007)	+0.040 (0.011)	−0.004 (0.007)

Standard errors in parentheses, in place of significance stars; see the note to Table 3.

In the financing rows atypicality does nothing at one stage and a great deal at the next. Whether a debut owner ever raises a Regulation D round — a private securities placement, reported to the SEC on Form D — is unrelated to how atypical its filing was: −0.017pp per standard deviation among owners who completed registration, against a 2.61% base, which is indistinguishable from zero ($t = -0.8$). Among owners who did raise one, the same measure predicts *reaching* the public market: +0.49pp on a 3.49% base, about 14% relative ($t = 3.8$). Atypical language is invisible at the financing stage and informative afterwards, which is what an investor screen that does not read the text would look like — the text being free and public at the moment of filing. The comparison is among survivors, so it cannot separate that reading from selection on unobservables; and the null at entry is a null, not evidence of indifference.[5]

That last row turns on one construction. An exchange-registration marker on its own does not mean a firm listed *after* raising: 56% of matched owners carrying one registered before their first Regulation D notice, because established public companies also place securities privately. The row is therefore restricted to owners whose exchange registration postdates their first notice, which halves the coefficient and leaves it significant.

Three limits bound the financing rows. Regulation D notices are electronically filed only from 2009, so they rest on 2009–2018 debut cohorts. Owners are matched by normalized name, which succeeds for a small and unrepresentative minority favouring formally constituted entities, so every estimate is conditional on matchability. And the final row conditions on funding, realized after the filing being scored, so the coefficient is a contrast among survivors rather than a treatment effect: it establishes that information distinguishing eventual listers was already present in the text years earlier, not that the language caused the outcome.

[5] This pair is sensitive to the reference window. Under annual reference buckets the entry row was −0.06pp ($t = -2.8$), which read as active negative selection. Per-filing windows move it to zero and leave the listing row intact, so the negative selection was an artifact and the informativeness among the funded was not.

Lead, by contrast, does not reverse anywhere after registration. It is negative at both use-proof gates, at financing with or without registration imposed, and at listing among the funded; the only row where it is not negative is SEC reporting among registered owners, where it is indistinguishable from zero (−0.004pp, SE 0.007). Being early is penalized at every stage where the market, rather than the examiner, is doing the selecting, and never rewarded at any of them.

5 Business vocabulary flows out of software, and entrants do not carry it disproportionately

The spatial version of this question is answered: von Graevenitz et al. (2022) show that novel description tokens reach distant US locations more slowly than near ones, in the *Regional Studies* programme surveyed by Castaldi & Mendonça (2022). The cross-industry version is open, and the Nice class system supplies the partition.

For nine curated business-model phrases, Table 6 records the first year each of four classes (software 009, tech services 042, transport 039, advertising/retail 035; 2.32 million granted filings) crosses a 1% prevalence floor. Software- and tech-services-origin phrases arrive in the physical-economy classes with lags of one to thirteen years; the only counterexample in the set is on-demand, which originates in transport. The 1% floor is sensitive to class size, and the asymmetry is a four-class result a 45-class build would test properly.

Table 6: First year a theme reaches 1% prevalence among granted filings, by class. "–" = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	**2009**	2013	2012
mobile-app	2010	**2009**	2012	2012
cloud	2012	**2010**	2011	2014
streaming	**2008**	2008	2013	2014
artificial-intelligence	2017	**2016**	2018	2018
augmented/virtual-reality	**2015**	2016	2022	2022
blockchain / crypto	2021	**2018**	–	2022
real-time	**2001**	2003	2009	2014
on-demand	2017	2016	**2014**	2017

Unsupervised theme extraction corroborates the pattern without depending on which phrases were picked. A detection-purpose latent Dirichlet allocation (LDA) topic model ($T = 50$ on a stratified 200,000-filing sample) yields recognizable business-model components; for each (theme, class) cell crossing the 1% floor with six or more tracked years, Bass-model fits (Bass, 1969) on 118 retained cells give median innovation coefficient $p = 0.018$, median imitation coefficient $q = 0.110$, and median $q/p \approx 6.5$, in the range typically reported for consumer-product adoption. Tagging each multi-class theme's origin class and counting directed origin-to-arrival edges: software contributes 33 of 72 edges against 18 expected under symmetry, and receives only 10. The asymmetry reproduces under independently-fitted non-negative matrix factorization (NMF) on the same sample; both are bag-of-words factorizations, so the robustness is within-paradigm.

Entrants carry the canonically new themes, but no more often than they carry anything else. The 50 earliest carriers per theme are 73.7% entrants on average — but the corpus's year-matched debut baseline is 76.4%, so the theme-carrier excess is −2.7pp and 28 of 50 themes sit *below* baseline. The themes leaning incumbent include environmental/sustainability, natural gas and petroleum, artificial intelligence, and cloud computing. So the arrival of new vocabulary is not an entrant phenomenon in the way the Schumpeterian reading would predict (Schumpeter, 1942).

The AI era has so far compounded rather than erupted, though the most recent data test that reading rather than confirming it. Aligning era starts (internet 1994, AI 2015), the two track closely through year three, then diverge: internet vocabulary tripled between years four and six to 23% of filings in 2000, while AI vocabulary reaches 12.9% at year nine, roughly where the internet stood at year five (Figure 6). The last two observed years are the steepest in the series — prevalence rises about 14% a year to 2021, then 50% between 2022 and 2023 and 26% into 2024, following the release of consumer chatbots. That is an acceleration rather than a discontinuity, since per-year corpus turbulence shows no AI-era deviation through 2024, but it runs in the direction the eruption reading predicts. The comparison is sensitive to the era-start alignment, and the falsifiable marker is close to binding: if AI follows the internet's path with a lag, its prevalence should triple within roughly three years of the 2024 edge and the entrant share of its carriers should rise above baseline.

6 Lead tracks commercial risk; atypicality does not

Vocabulary position carries two separable signals, and it is the signed one that tracks what becomes of a product. Lead marks commercial risk: +1.4pp top-versus-bottom quintile at the use-proof gate with fixed effects, controls, and owner clustering, and -0.047pp per σ in SEC reporting at equal atypicality, so the filings whose language an industry went on to adopt belong disproportionately to products that do not survive and to owners who do not reach public reporting. Atypicality runs the other way and on a different margin: it marks marks that survive their gate (+1.3pp per standard deviation), and its quintile contrast on SEC reporting is zero to three decimals once class, year and length are absorbed (Table 4). A continuous per-standard-deviation term on the harmonized sample of Table 5 does return +0.042pp on a 0.56% base; that is the one specification in which it survives, and it is a linear term through a non-monotone profile.

ΔKL, adapted from Barron et al. (2018); Murdock et al. (2017) onto topic distributions of USPTO trademark filings, produces interpretable structure on commercially-incentivised legal text. The construct is empirically distinct from patent-track invention within firm: no stable association survives a change of text representation, and no estimate reaches a tenth of a standard deviation. And the corpus supports direct measurement of cross-industry vocabulary diffusion, with multi-year origin-to-arrival lags, Bass-fittable adoption, and asymmetric flow out of software on the four classes built so far.

ΔKL does not measure innovation in any strong sense; the defended primitive is vocabulary position on commercial text, and Appendix G shows concretely which kinds of novelty the primary scoring cannot see. ΔKL does not identify an exploitable extreme-tail return premium; the large debut-specification figure fails a sample-size diagnostic. And the term-scored signed firm-level advantages of leading filings (margins, listing) are artifacts of drafting style and lexical specificity (Appendix C).

6.1 Five neighbouring constructs this is not

Novelty and *transience*, in the sense of Murdock et al. (2017) and Barron et al. (2018), are the two KL levels themselves: novelty is divergence from the past, transience divergence from the future. Atypicality as used here is their average, and so is neither of them alone; *novelty* in particular is declined because it names only the backward-looking half of a quantity that faces both ways. *Resonance* is their signed difference, and is arithmetically what this paper calls lead. The word is avoided because it names a mechanism — the field resonating with an author's contribution — that this design cannot observe. A filing can sit where its class was heading because it was imitated, because it and the class responded to the same outside event, or by coincidence; the measure does not separate these, and *resonance* would assert the first.

Innovation is a property of the product, not of the sentence describing it: two firms selling the same thing can draft very differently, and a genuinely new product can be described in boilerplate (Appendix G). The measure sees prose.

Radicalness (Dewar & Dutton, 1986; Henderson & Clark, 1990; Tushman & Anderson, 1986) is the closest neighbour, since KL is a departure measure rather than a newness count and the Amazon exhibit is a large departure built from unremarkable components. It is declined because it claims that existing competences are devalued, and nothing here licenses the step from unusual language to devalued competence.

Distinctiveness has two prior occupants. In trademark law it is the registrability doctrine, proven in part by the continued use this paper uses as an outcome. In strategic management, optimal distinctiveness (Zhao et al., 2017) predicts an inverted-U in conformity — a shape that appears here at registration but not at listing, so the term would imply a theory is being tested when it is not.

What *atypicality* says is narrower than any of them: this description is unusual for its class and year. Whether the thing described was new, radical, or good is not measured.

6.2 Filings are not independent draws

Some of them share an exact position with another filing, and outcomes cluster within owner — the within-owner correlation is 0.42 for gate survival and is mechanically 1.0 for public listing, since listing is a property of the owner and a firm with hundreds of marks contributes hundreds of identical successes. The estimates above are unaffected, because the gate regressions cluster on normalized owner and the firm-level specifications carry one observation per owner. But binning filings without that correction overstates how much structure the corpus contains: on a positional grid, correcting cell errors for the clustering design effect — the factor by which within-owner correlation inflates the variance of a cell mean — cuts apparent excess dispersion by roughly a quarter on the gate outcome and by half on listing. Displays of the vocabulary plane are descriptive here for that reason, and the near-zero pooled correlation between A and L reported in Section 3 does not survive cell by cell within class, so per-cell contrasts should not be read as within-class effects.

6.3 What remains unresolved

What drives the episodic pattern is not settled. Cohort and gate year move together, so post-crisis bet composition, app-era speculative filings, and calendar-time enforcement changes cannot be separated on this design. The representation is also fit on the pooled corpus; a vintage refit would close the remaining look-ahead channel, though the months-scale freshness variant is already look-ahead-free and reproduces the risk pattern.

7 Reproducibility

Code, firm-year panels, and analysis outputs are at `https://github.com/ericsilver/tm-vocabulary`, indexed in the repository `README.md`. The pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command (the TRTYRAP backfile is about 12 GB, streamed once). Published panels include the SEC-matched firm-year ΔKL panel (n = 59,033) and the PatentsView-joined panel (n = 174,569). An online appendix under `docs/online-appendix` carries a three-panel breakout for each of the 45 Nice classes, the cross-industry comparisons, and a machine-readable table of per-class contrasts with standard errors.

A Computation, corpus, and robustness

A.1 Corpus and scoring

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work uses the complete software/electronics class (Nice 009; 1.81M filings); the diffusion application uses four classes totalling 2.32 million granted filings 1990–2024; the gate analyses use all 45 classes. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees by normalized-name join.

The topic representation is a $T = 50$ LDA fit on a stratified sample of 448,437 filings across all 45 classes (vocabulary of 62,168 unigrams and bigrams); every filing 1990–2024 is transformed to a dense 50-topic distribution, and prospective and retrospective KL are computed against per-filing topic references spanning 1,826 days on each side of the filing date, scoring 1995–2019. A filing whose window holds fewer than 500 same-class filings on either side is left unscored, since a thin reference inflates the divergence mechanically; that floor, together with the corpus edges, is why roughly two-thirds of filings carry a score. A term-level variant is retained alongside, on the same anchoring: token distributions (within-class document frequency ≥ 50) against per-filing windows of the same 1,826 days, Dirichlet-smoothed ($\alpha = 1$) over the class vocabulary. Because a goods/services description runs to a few dozen words, the divergence is needed only at the terms a filing actually uses, so the reference is carried as a running count vector and read sparsely. The term variant is used where vocabulary identity is the object (phrase transit, the window decomposition, the freshness measure) and as the comparison scoring throughout. Both representations are also computed under an exponentially decayed reference with a two-year half-life, truncated at the same window; Appendix A.4 reports what changes.

A.2 Scores are interpretable from 1995

Burn-in is the span at each end of the corpus where the reference windows cannot be filled, so scores there reflect a thin reference rather than the text being scored.

The signal depends on having enough past and future filings to populate the references; the 5-year past window is incomplete before 1990 and the future window after 2020, and both zones are excluded (Figure 7). The burn-in cut is set empirically: thin past references inflate prospective surprise, and profiling that bias across all 44 sufficient-volume classes gives a median absolute bias of ∼2.0 nats — the natural-log unit in which these divergences are measured — in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any interpreted effect size. The 1995–1997 deviations (∼0.15–0.17) are not burn-in — mechanical contamination cannot rebound — but the dot-com era moving the corpus. Per-class burn-in lengths estimated under a stability-band criterion track each class's noise floor rather than its reference density, so a single standard cut at 1995 is used, conservative by about two years.

A.3 Lead is a small residual, and term scoring tracks length

Past-facing and future-facing KL are highly correlated however scored. On the complete class-009 corpus under the production scoring they correlate at $r = +0.988$ ($n = 1{,}109{,}643$); under term scoring on class-year references the figure is +0.971. The lead L is therefore a small residual of two large, stable quantities: both levels have a standard deviation near 0.70 nats about a mean of 1.46, while L has a standard deviation of 0.109 — about a sixth of either level's spread. Only 1.2% of the two levels' combined variance survives into their difference. That is what makes L sensitive to anything that inflates the levels, and under term scoring the dominant such thing is document length. Term-scored atypicality correlates with log filing length at -0.729 in class 009 and -0.688 in class 035: the shorter the description, the spikier its term distribution and the further it sits from any smoothed reference, so "unusual" and "short" are very nearly the same measurement. Topic scoring breaks that identity, taking the same correlations to -0.251 and -0.125. The consequence for the lead is direct: the standard deviation of L rises threefold across deciles of A under term scoring and by a quarter to a half under topic scoring, so a unit of lead means roughly the same thing everywhere only in the topic representation. Appendix F shows this as a length surface over the two axes. Topic scoring returns a finite value for about 65% of filings, so the two columns are not the same sample; re-estimating the term-scored correlations on exactly the topic-scored subset still gives -0.722 and -0.692, which makes the contrast a property of the representation rather than of the sample, but the like-for-like spread comparison is closer to twofold against 1.3–1.6-fold than to the headline threefold. And "flat" means no monotone length gradient rather than constant: class 035 under topic scoring is mildly U-shaped in A. The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.36 on 1.9M jointly scored observations), which gives their agreement on the mark-level patterns evidential force and their disagreement on the firm-level patterns diagnostic force (Appendix C). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT "most innovative" lists sit $+0.09\sigma$ above the panel mean under term scoring ($p = 0.22$), $+0.14\sigma$ under topic scoring ($p = 0.07$), and $+0.08\sigma$ at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with ΔKL on the same panel (Pearson +0.010).

A.4 The mark-level results hold under every scoring

The snapshot cross-check on the single one-gate-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: -5.4pp survival across quintiles under topic scoring and -5.7pp under term scoring on the identical sample (Figure 8). Under term scoring, gate survival *rises* in the raw KL level (+6.5pp prospective, +7.6pp retrospective): atypical text survives the use requirement better while leading text survives it worse — the same two-axis separation as Section 4.4, at the mark level.

A.5 Annual reference buckets overstate the penalty by a third

Every estimate uses a reference anchored on the filing's own date. The alternative is one reference per class-year, under which a filing made on 1 January and one made on 31 December receive identical references and neither sees any of the language filed during its own year. Appendix B reports the calendar signature that leaves.

First, the gate penalty was overstated by about a third. On the same 3.10 million registrations, the top-versus-bottom contrast is +2.15pp (SE 0.30) under annual buckets and +1.39pp (SE 0.17) under per-filing windows. The attenuation is remarkably uniform across specifications — the ratio runs 0.59 to 0.73 for every sample split in Table 3 — and it is an attenuation of magnitude only: standard errors fall by roughly half, so the t-statistic *rises*, from 7.2 to 8.3. The same holds in the software class alone, where the contrast moves from +7.40pp to +5.25pp.

Second, the shape of the gate penalty changes. Under annual buckets the quintile profile looked like a gradient with a steeper final step; under per-filing windows it is close to a threshold, with the top quintile carrying about seventy percent of the total and the interior quintiles nearly flat. The coarser measure spread a top-quintile effect across the whole range.

Third, one published shape does not survive. Registration completion showed an inverse-U on both the unsigned and the signed axis. Only the unsigned one is real; on ΔKL the curvature disappears entirely under per-filing windows, and the earlier interior peak is attributable to the month gradient above. The firm-level results are the least affected by the window change, though they are sensitive to the owner-to-registrant linkage instead (Section 4.4): the signed decomposition keeps both its signs and its significance under either window.

A.6 Signal is flat between five and seven years

The window is a parameter of the measure, not a property of it, and the five years used throughout was inherited from the class-year construction rather than chosen under the current one. Re-scoring the software class at several widths under per-filing references, and judging each by the first-gate quintile contrast it produces, gives +2.02pp at two years, +2.20 at three, +3.23 at five and +3.37 at seven, each on an identical sample of 452,911 registrations and each with a standard error of 0.23pp. Signal rises steeply from two to five years and is flat between five and seven: the +1.03pp gain from three to five is well outside the sampling error, the +0.14pp from five to seven is not. A ten-year window returns +2.89pp, but that figure is not comparable to the others — requiring both windows to lie inside the corpus costs 22% of the scored filings and 13% of the gate sample, dropping the earliest and latest registration years, which are also the years whose contrasts differ most. The apparent decline at ten cannot be separated from that change in composition and is not evidence about window width. On the range where the sample is fixed, seven years is nominally the peak and five is within noise of it while retaining more of the corpus, so five is defensible rather than optimal. The sweep is on one class and one outcome.

A.7 The gate punishes breaking with the past, not resembling the future

"Leading" is relative to a reference, and mixing windows across the two sides identifies which novelty the gate punishes. Scoring the corpus at symmetric windows of $W \in \{3, 5, 7\}$ *years* — this appendix measures the window in years, where the main text measures it in days — plus a foresight configuration (prospective against the recent 3-year past, retrospective against the distant 7-year future) and a past-rupture configuration (the reverse): the top-versus-bottom gate lift runs −3.2pp for foresight, −4.7pp at symmetric $W = 3$, −5.7pp at the $W = 5$ baseline, −6.1pp at $W = 7$, and −7.3pp for past-rupture, the most pervasive variant (negative in 35 of 41 classes; Figure 9). In software the decomposition is complete: foresight carries no penalty (−0.3pp) while past-rupture carries −12.7pp. Use-proof-gate risk attaches to breaking with the established past, not to resembling the future, consistent with vocabulary leaving the corpus gradually: the long-past reference holds the fading establishment, and distance from it is what the use requirement punishes. The registration inverse-U is indifferent to the window mix (+4.6 to +4.9pp depth across all five variants), so the examiner and the marketplace respond to different aspects of the same text.

The decomposition extends below the year. Scoring class 009 monthly against two adjacent references (the twelve months ending one month before filing, and the twelve before those), the two levels are nearly identical per filing ($r = 0.992$) but their difference — a months-scale freshness measure, high when a filing resembles the very recent months far more than the year before them

— reproduces the risk pattern at −9.4pp across quintiles, stronger than the symmetric baseline in the same class, with the penalty on the high-freshness tail — filings whose language runs ahead of the recent months rather than behind them — which is the same direction as the annual-window result. Freshness is built entirely from pre-filing text and is computable on the filing date with no look-ahead; corpus-wide estimation is owed.

B Representation and resolution drive the score; weighting does not

B.1 Representation, anchoring, weighting: which of the three matters

A scoring is fixed by two choices. The *representation* is what the language is turned into: a distribution over 50 fitted topics, or a distribution over the class vocabulary's terms. The *reference* is what that is compared against, and it hides two sub-choices — the *anchoring* (one reference per class-year, or one per filing built on its own date) and the *weighting* inside the window (every day equal, or decaying by half every two years). Six builds exist:

Representation	Anchoring	Weighting	sd(lead)	sd(atypicality)
Topics	class-year	flat	0.132	0.672
Topics	per-filing	flat	0.102	0.698
Topics	per-filing	half-life 2y	0.082	0.698
Terms	class-year	half-life 2y	0.299	1.330
Terms	per-filing	flat	0.155	1.101
Terms	per-filing	half-life 2y	0.130	1.102

All six are compared on the 6.01 million filings every one of them scored. The second row is the production scoring. The fourth is the legacy build, and it is the one to handle carefully: it moves anchoring *and* weighting at once, so a result computed on it cannot be attributed to either. That is why the two per-filing term builds were constructed.

Changing one choice at a time settles the ordering. Correlations of lead between builds differing in exactly one respect:

Choice varied	r (lead)	r (atypicality)
Weighting, flat vs half-life 2y (topics)	0.982	1.000
Weighting, flat vs half-life 2y (terms)	0.970	1.000
Anchoring, class-year vs per-filing (topics)	0.738	0.929
Resolution, 50 vs 200 themes	0.636	0.828
Representation, topics vs terms	0.460	0.396

The weighting is close to irrelevant: whether recent language counts for more than distant language within the window barely moves the score, and it does not move atypicality at all to three decimals, which is what the 45-degree rotation of Section 3 implies — the long axis is a property of the filing and the weighting acts almost entirely on the small residual. Anchoring matters moderately. Resolution and representation matter most: quadrupling the number of themes leaves the two builds agreeing on 40% of the variance in lead, and topics against terms on less than a quarter. A robustness check that varies only the weighting is therefore close to a restatement; one that varies the representation or the resolution is a real test, which is why the term-scored replication in Appendix A.4 and the sweep below are the ones the claims lean on.

B.2 Fifty themes is coarse, and the finding does not depend on it

Fifty themes over a 62,168-term vocabulary is roughly one theme per industry, so the partition is coarse enough that a reader may reasonably suspect the findings belong to it. Raising the theme count does not add coverage — the vocabulary is fixed, and a larger T re-partitions the same words — but it changes what counts as sitting far from an industry, and could change any sign. Re-fitting the model at four resolutions on the same stratified sample and the same analyzer, and re-scoring the software class under per-filing windows:

Themes	Gate contrast, lead	t	Gate contrast, atypicality	r with $T = 50$
50	+3.23pp	13.8	−14.01pp	—
100	+2.71pp	11.6	−13.69pp	0.690
200	+5.20pp	22.2	−13.45pp	0.640
500	+5.69pp	24.3	−14.49pp	0.641

Three readings, in descending order of how much weight they carry. The sign and the significance are untouched across a tenfold change in resolution, and so is atypicality, which stays within a point of −14pp throughout. The magnitude is not: the lead contrast runs from +2.71 to +5.69pp and does not move monotonically in T, so no single resolution can be quoted as the software class's effect, and the figure this paper reports at $T = 50$ sits at the low end of that range rather than the high one. And agreement with $T = 50$ settles at 0.64 without decaying as T grows, which says the resolutions share a fixed core and differ in the remainder rather than drifting apart as the partition refines.

The corpus-wide contrast is steadier than the single class: at $T = 200$, scored for all 45 classes, it is +2.23pp against +2.29pp at $T = 50$. Only those two resolutions exist corpus-wide, so that is a two-point comparison, but it is the one the paper's estimates run on, and pooling evidently averages away much of the per-class resolution sensitivity.

Annual anchoring leaves a signature that per-filing anchoring does not. Mean lead by calendar month of filing, in units of the build's own standard deviation, has no reason to trend — nothing about filing in December rather than January makes a description more forward-looking:

Representation	Anchoring	December − January (sd)
Terms	class-year	+0.169
Topics	class-year	+0.041
Topics	per-filing	−0.008
Terms	per-filing	+0.017

A December filing sits eleven months further from its past reference and eleven months nearer its future one, and under annual buckets that shows up as position in the calendar. Under per-filing references it is gone.

The substantive claim does not depend on any of this. The first-gate contrast, computed on the common sample, runs +2.73, +2.71, +2.39, +4.05, +3.97 and +3.19pp on lead across the six builds in the order tabulated above, and −7.75 to −6.17pp on atypicality. Every sign holds; term scoring returns the larger lead contrast, so the headline estimate is the conservative representation.

B.3 One gate is linear; the public-reporting profile is a U

Four shapes were fitted to each profile by weighted least squares on 40 equal-count bins (Figure 10): linear, quadratic, an exponential $a + be^{cx}$, and a free-vertex symmetric form $a + be^{c|x-x_0|}$ meant to catch a U directly. Selection is by AIC.

Gate	Axis	R^2 linear	R^2 selected	ΔAIC	Selected
Registration	atypicality	0.38	0.92	77.1	symmetric, vertex +1.02
Registration	lead	0.08	0.17	1.9	quadratic
Use-proof	atypicality	0.74	0.74	0.0	linear
Use-proof	lead	0.44	0.55	4.3	symmetric
Listed	lead	0.19	0.47	12.8	symmetric, vertex +0.35
Listed	atypicality	0.74	0.83	11.5	symmetric, vertex +2.30
Reporting	lead	0.07	0.66	36.1	symmetric, vertex +0.26
Reporting	atypicality	0.64	0.80	18.4	symmetric, vertex +2.18

The free-vertex form wins in six of the eight profiles, but it wins in two different ways and the distinction matters. On the two atypicality profiles it is a genuine exponential decay away from a vertex, with fitted exponents of 0.49 and 0.85. On the lead profiles it degenerates: the fitted exponent is of order 10^{-4}, which drives $e^{c|x-x_0|}$ to $1 + c|x - x_0|$, so what actually wins there is a V — linear in distance from a free vertex — rather than anything exponential. A V beating a parabola on a U-shaped profile says the turn is sharper than a quadratic allows, not that the process is multiplicative.

The registration profile in atypicality is decisively curved: an inverted U peaking a standard deviation above the mean, with the fit more than doubling the explained variance. This is the middling-specificity pattern of Appendix D, and a linear summary of it is close to meaningless. The use-proof gate in atypicality is the opposite case — genuinely linear, with every curved form buying nothing — so the per-standard-deviation slope the paper quotes there is the right summary.

The public-reporting profiles are where the linear reading actively misleads. Both are U-shaped in lead, and the fitted linear slope runs *downward* through a curve whose two tails are its high points. Quoting only that slope would report that leading owners reach public markets less often, which is true on average and false at the extreme the sentence would be read as describing. The vertex sits above the mean in both, so the lagging tail is the longer arm of the U — the asymmetry Section 4.4 reports from the quintiles.

AIC here is computed on 40 binned means, not on the millions of underlying filings, so it ranks shapes rather than testing them. The 77 at registration and the 12 to 36 at the public-reporting gates are decisive; the 1.9 at registration on lead and the 4.3 at the use-proof gate are weak, and those two rows are better read as "no worse than linear" than as curvature.

B.4 Listing and merely reporting behave the same way

"Appears in SEC records" mixes two populations: companies whose shares trade, and entities that file with the Commission without ever listing — private issuers, funds, subsidiaries registering securities. Splitting the 19,889 matched owner strings on whether their CIK carries a ticker in the Commission's own company-ticker file gives 10,556 listed against 9,333 that file financial statements without one.

It does not change the finding. Among registered debut owners whose debut filing is scored, 0.310% later appear as listed companies and 0.285% as reporting non-listed ones, and both profiles are U-shaped in lead with vertices at +1.35 and +0.56 standard deviations. The listed profile is slightly flatter and noisier, which is what the smaller and more selected outcome predicts. Whatever the signed axis is picking up at this margin, it is not an artifact of counting non-operating registrants as commercial successes, and it is not specific to the act of listing either.[6]

[6] Two limits on the split. The ticker file is a current snapshot, so a company that listed in 2003 and was acquired in 2011 appears in the non-listed tier; the tiers are therefore "currently listed" rather than "ever listed," which will

C The measurement hazard: term scoring manufactures firm-level signal

Under term-level scoring, leading filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in SEC EDGAR rises in term-ΔKL from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched firm-year panel, trailing 3-year mean term-ΔKL predicts gross margin at +2.3pp per σ ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the published panel through 2024), with prospective surprise entering positively and retrospective negatively; and firm-year term-ΔKL predicts excess stock returns of +1.5 to +5.0pp per σ at one-to-four-year horizons. A larger debut-only return figure (+28pp at 4 years) fails a sample-size diagnostic and is not claimed.

None of the signed versions survives distribution scoring. On identical samples, the debut listing gradient in signed topic-ΔKL is flat to declining, and the margin coefficient reverses: −1.3pp per σ ($t = -1.8$) at $T = 50$, −2.2pp ($t = -3.1$) at $T = 200$, and −0.8pp ($t = -1.1$) at $T = 500$, against the term-level +2.3pp; the prospective/retrospective decomposition flips sign and stays flipped. The term-level listing gradient separately dissolves under document-length controls: within length bands it is U-shaped rather than rising, with the monotone rise contributed by composition across lengths (listed firms' counsel write longer descriptions). The mark-level results are unaffected — they hold under both scorings at every resolution.

Vocabulary blindness does not rescue the signed signal. The topic model's vocabulary is sample-built, so rare and emergent terms are invisible to it; if the firm signal lived exactly there, topic scoring would erase a real signal rather than an artifact. Computing each filing's invisible-vocabulary share (63.8% of class-009 term *types* but only ∼9% of token *mass*): the share itself is a strong firm marker (P(EDGAR) 2.9% to 5.5% across its quintiles; gate survival +5.7pp), but *within* invisible-share strata the signed ΔKL gradient on the firm outcome is flat to negative, while the gate penalty holds in both strata (−6.9 and −8.3pp). The term-level firm signal is *unsigned specificity*: serious firms describe specific products with specific, therefore rare, words, and rare-term mass inflates term-ΔKL mechanically. Topic count is a partition dial, not a coverage dial — a crypto/blockchain topic exists at $T = 200$ but not at $T = 50$ or $T = 500$, because raising T re-partitions the fixed vocabulary along dominant axes — so resolution sweeps test granularity, not coverage, and the invisible-share split is the test that binds.

The registration filter does not explain the divergence. Dropping the registration conditioning leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), and an examiner channel touching 4.3% of failed applications cannot generate several-point quintile gaps.

Term-resolved text-novelty measures can therefore manufacture firm-performance correlations from drafting style and lexical specificity, and a distribution-based rescoring is a cheap, decisive diagnostic.

D Examination rewards a middling specificity

Across 3.59 million debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure 2, panel C). The same shape appears on the past-facing level alone,

misclassify older successes downward. And the base rates here are computed on the appendix's own panel, which conditions on the debut filing carrying a score and so is smaller than the body's sample.

more mildly, running 54.0% at the bottom quintile to 58.4–59.1% in the middle and 57.7% at the top (Figure 1A).

The signed axis behaves differently. Completion on ΔKL falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end. That trough is composition rather than timing: $|L|$ correlates with atypicality at +0.31, so filings near zero lead are disproportionately low-atypicality filings, which complete poorly for the reason above. Split by atypicality, the trough is absent in the upper three fifths and survives in the lowest. Reference windows built by calendar year put an inverse-U on this axis too, but that one is an artifact of annual bucketing (Appendix A.5) and does not survive at daily resolution. The signed axis still enters a linear specification strongly (Table 5), but monotonically rather than through an interior peak: what curvature exists at registration belongs to how far a filing's language sits from its category, not to which side of it the filing sat.

The obvious objection is that unfamiliar language is simply refused more often, so the measure recovers nothing beyond familiarity. Completion falls at *both* ends, so the most category-typical filings also complete less than the middle, which familiarity cannot explain; and the curvature is specific to the unsigned axis, the signed axis showing no interior peak at all.

Refiling after abandonment is itself mildly related to vocabulary position, though not on the axis that matters here. Refiling rates are flat in atypicality (a middle-minus-tails difference of −0.06pp on a 3.6% base) but rise with lead, from 2.6% among the most lagging abandoned applications to 3.4% among the most leading. Firms whose language ran ahead of their industry are slightly likelier to try the same mark again.

E The outcome analysis without registration conditioning

Figure 11 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by ∼2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is nearly flat in ΔKL — 22.4% at the bottom quintile against 22.1% at the top, a middle-versus-tails depth of +0.001pp (SE 0.10) — because two opposed gradients cancel: registration rises in lead (61.3% to 63.2%) while gate survival falls (44.2% to 40.8%). Conditioning on grant is what separates them. Panel B shows the conditional gate outcome on the same filings, and it is U-shaped rather than inverse-U (depth −2.11pp), with a clearly penalised leading tail consistent with the cleaner single-gate cohort of Figure 8.

F Term scoring measures length; theme scoring does not

Figure 12 puts the two representations side by side over the same axes, with each hexagon coloured by the mean length of the filings in it. Under term scoring the colour runs from dark near the origin — a median of about 200 terms — to near-white at the far end, where the median is eight. That gradient is the measurement problem: a filing scores as atypical largely because it is short. Under topic scoring the same surface is a narrow band of roughly uniform tone.

Two filings are marked in both columns. Under the production scoring they behave differently from one another, which is the point. AMAZON.COM (1997) is a leader on both representations — the 98th percentile of its class on terms and the 94th on themes — while sitting at the 40th and 47th percentiles of atypicality, so both agree that it was early and neither mistakes it for strange.

Apple's iPod (2001) divides them: the 31st percentile of lead on terms against the 62nd on themes. The surfaces plotted here are on the class-year build the figure was generated from, where the length gradient is starkest; the percentiles just quoted are on the per-filing build used for every estimate.

Topic scoring is independent of document length, which term scoring is not; term scoring resolves novel word *combinations*, which topic scoring does not. Hence topic scoring for every estimate, since the length confound contaminates firm-level comparisons, and term scoring for every worked example, since the topic basis has no coordinate for the language the examples turn on.

G How three filings encode

Table 7 traces three filings through term scoring and topic scoring at $T \in \{50, 200, 500\}$ — three kinds of novelty, and how each representation handles them.

Amazon.com (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention ("computerized line," "line search," "search ordering"; class document frequencies 379, 134, 355). The filing's text reads "on line" as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword "on" and the surviving bigram "line search" is the act of forming the compound caught in progress. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under topic scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

Kombucha (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in the topic vocabulary but, with no kombucha-like topic, its mass dissolves into a generic beverages topic (≥ 0.60 weight at every T).

Ethereum (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services topics at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain topic — the eleven occurrences of *software* outvote the blockchain terms.

Table 7: Top topic loading for three filings, by representation. Term scoring resolves the novel content of all three; topic scoring resolves none of it at any tested resolution.

	Amazon.com (1997)	Kombucha (1997)	Ethereum (2018)
novel content (term)	line search, search ordering	kombucha, saccharose	blockchains, distributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, development (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)

The pattern across the row is the measurement point in miniature: topic scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting topic exists. That the mark-level outcome results are nearly identical under both scorings despite this blindness is what licenses topic scoring as a robustness instrument; that the firm-level results diverge is what exposes them as artifacts of the lexical specificity term scoring additionally captures (Appendix C).

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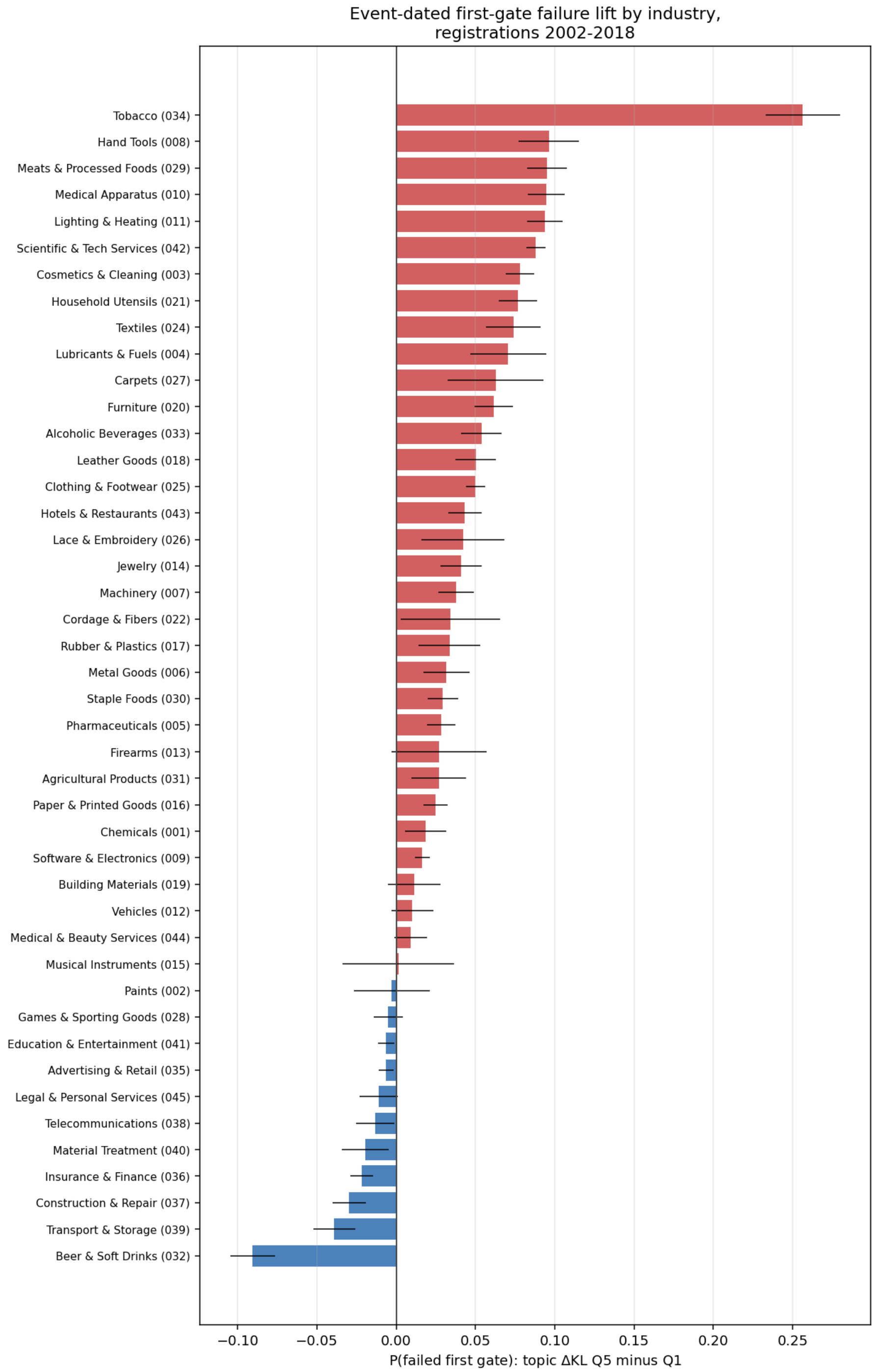

Figure 3: Event-dated first-gate failure lift (topic ΔKL top minus bottom quintile) by industry, registrations 2002–2018, raw quintile contrasts. Positive bars indicate the leading penalty. Whiskers are two-sample binomial intervals, uncorrected for the within-owner correlation (0.42) and so narrower than clustered intervals; read Table 3 for precision. Per-class breakouts are in the online appendix.

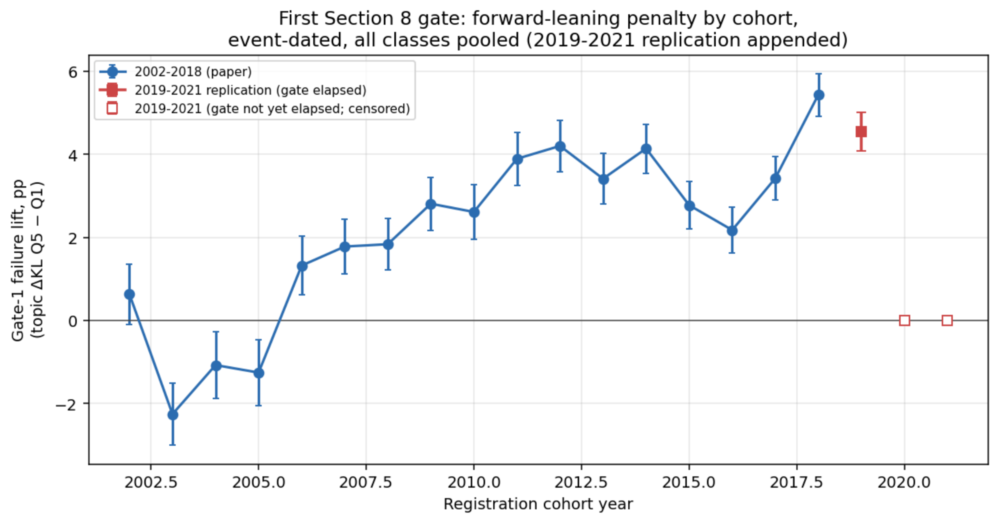

Figure 4: The leading gate penalty by registration cohort, event-dated, all classes pooled, raw quintile contrasts. The penalty is absent or negative for the 2002–2005 cohorts, turns positive in 2006, and is present in every cohort from 2008 through 2019, ranging +1.8 to +5.6pp with no downward trend. 2019 is the last evaluable cohort: cancellations post at about age six and a half and the record ends in April 2026, so later cohorts have not reached their gate rather than having passed it.

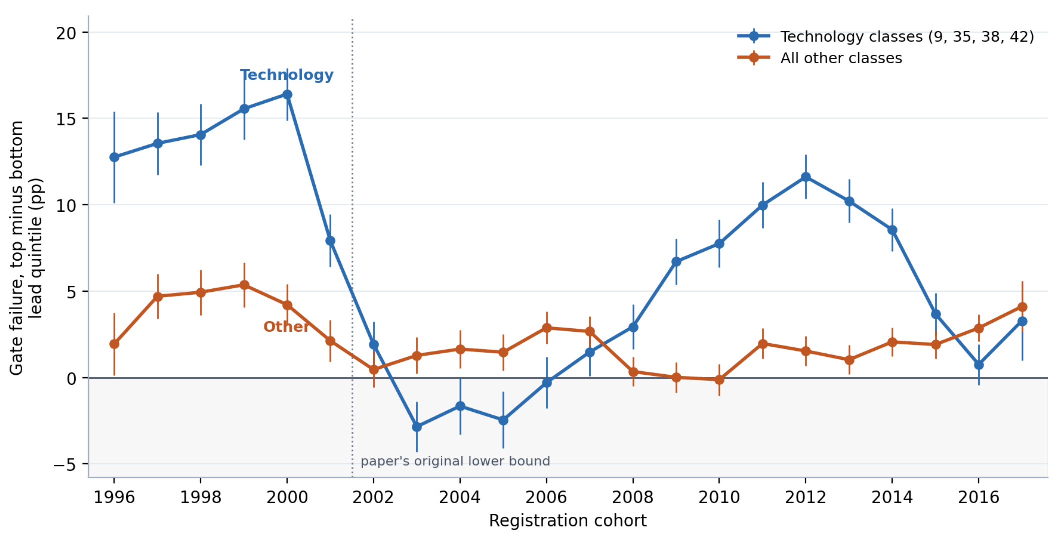

Figure 5: The leading gate penalty by registration cohort, technology classes against all others, raw quintile contrasts with 95% intervals. Cohorts are restricted to registrations whose ninth anniversary is inside the record, so every outcome shown is settled. Outside technology the penalty is small and almost flat across two decades; inside technology it swings from +16pp for the 2000 cohort to −3pp for 2003 and back to +12pp for 2012. The dotted line marks the 2002 cut that the main cohort series uses.

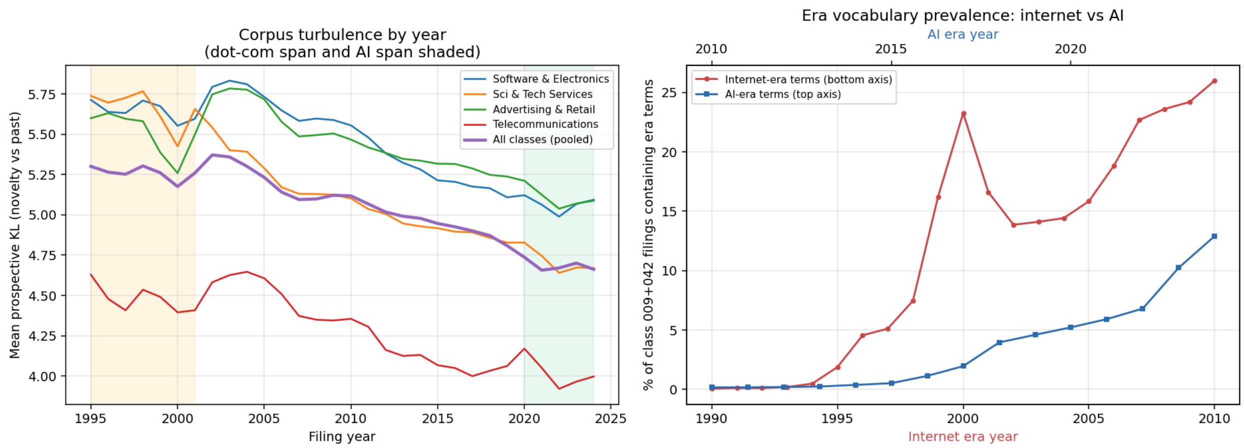

Figure 6: *Left:* per-year mean prospective KL by class and pooled, 1995–2024, with the dot-com and AI spans shaded. *Right:* era-term prevalence in classes 009+042, internet era (bottom axis, from 1994) against AI era (top axis, from 2015).

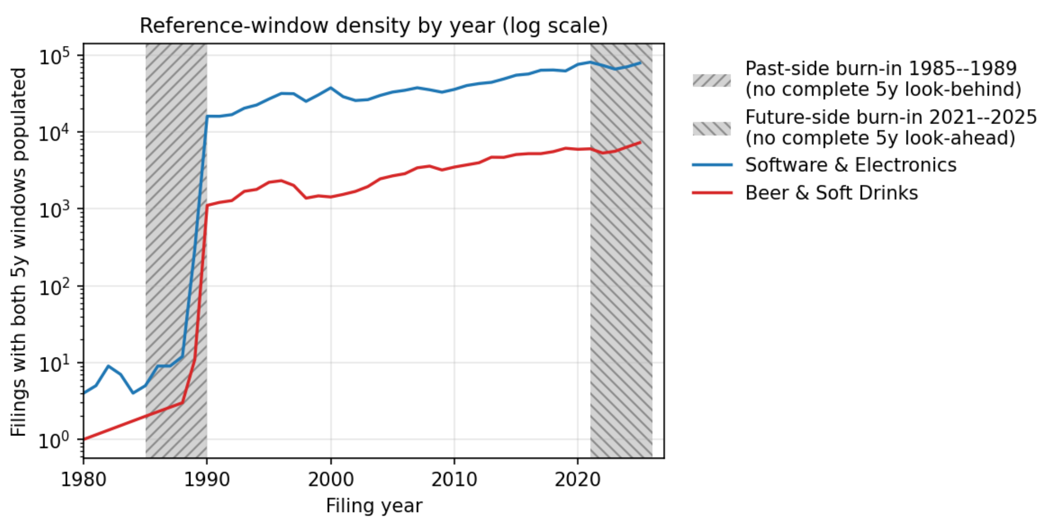

Figure 7: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete.

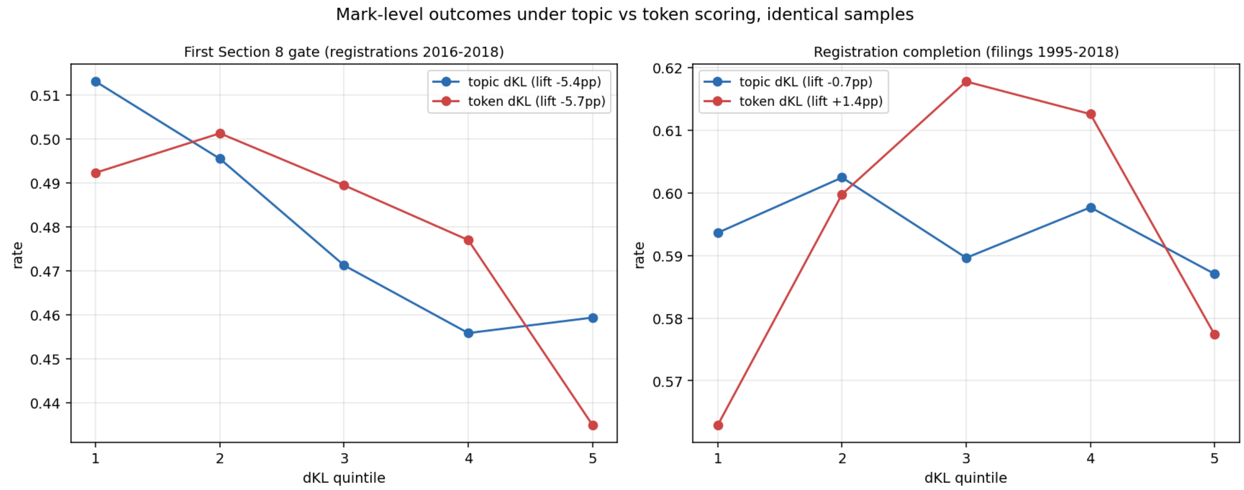

Figure 8: The two mark-level outcomes under topic and term scoring on identical samples, pooled across classes. *Left:* first use-proof gate survival (snapshot, registrations 2016–2018), where the two representations agree closely. *Right:* registration completion (filings 1995–2018), where they do not: the inverse-U in the signed axis is present under term scoring (depth $+0.048$) and absent under topic scoring (depth -0.001). The gate result is scoring-robust; the registration curvature on this axis is not, and Section 4 reports only the unsigned version.

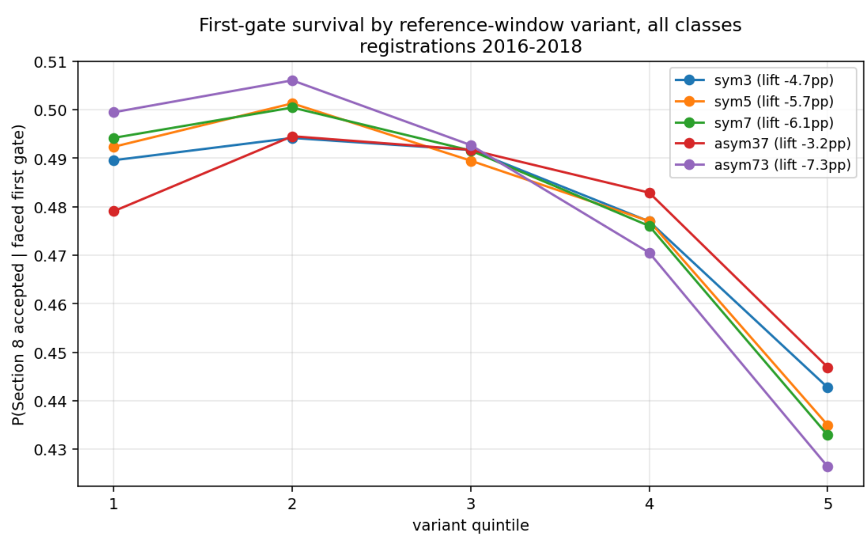

Figure 9: First-gate survival by quintile of each reference-window variant, pooled across classes, registrations 2016–2018. The penalty deepens monotonically from the foresight configuration (pros W=3, retr W=7) to the past-rupture configuration (pros W=7, retr W=3).

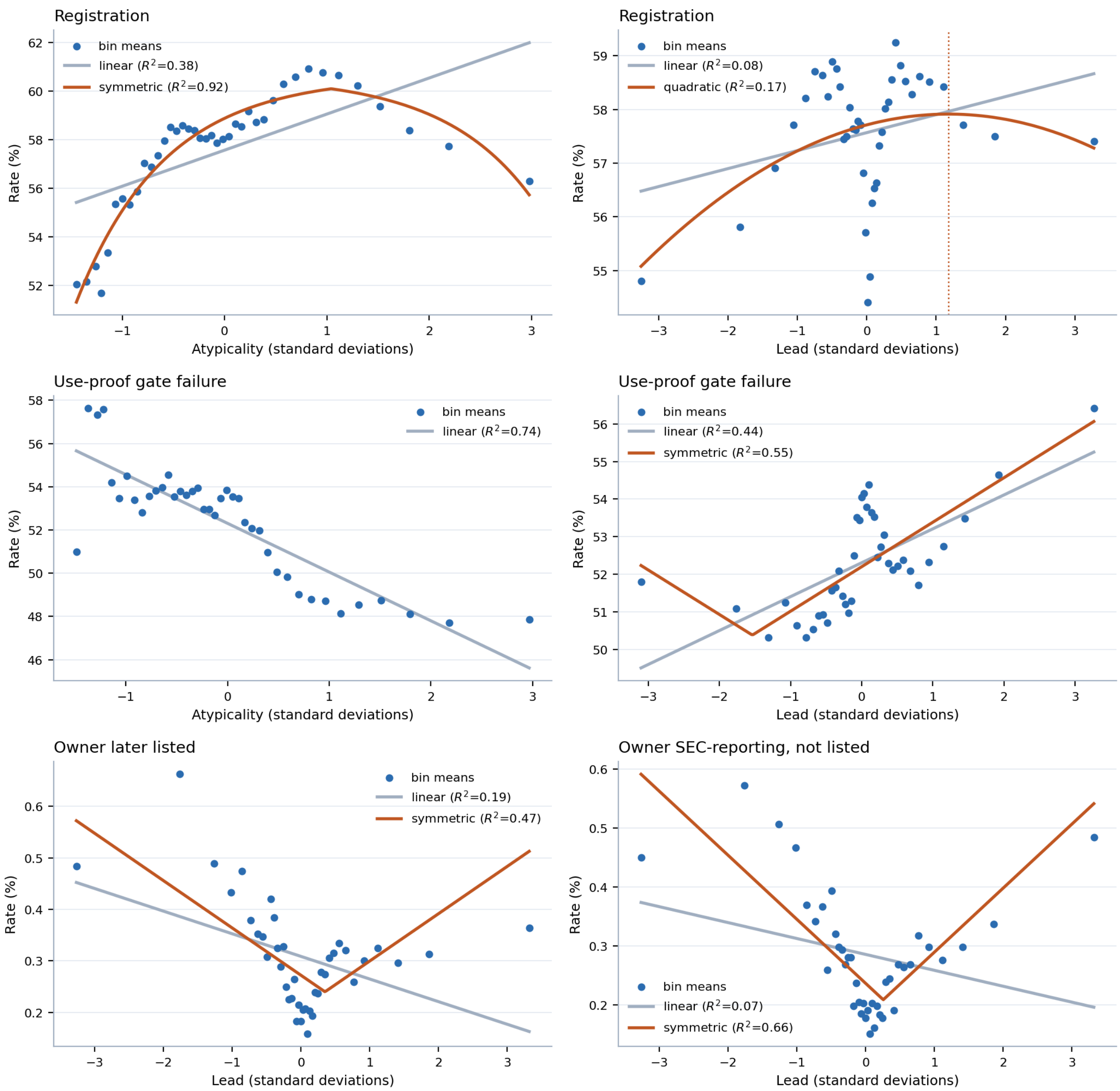

Figure 10: Binned profiles at the three gates, with the linear reading the paper's contrasts assume and the shape selected by AIC where that is not linear. Forty equal-count bins; the dotted line marks the fitted vertex. The bottom row is the case that matters: a linear fit through the public-reporting profile slopes downward while the data turn up at both tails.

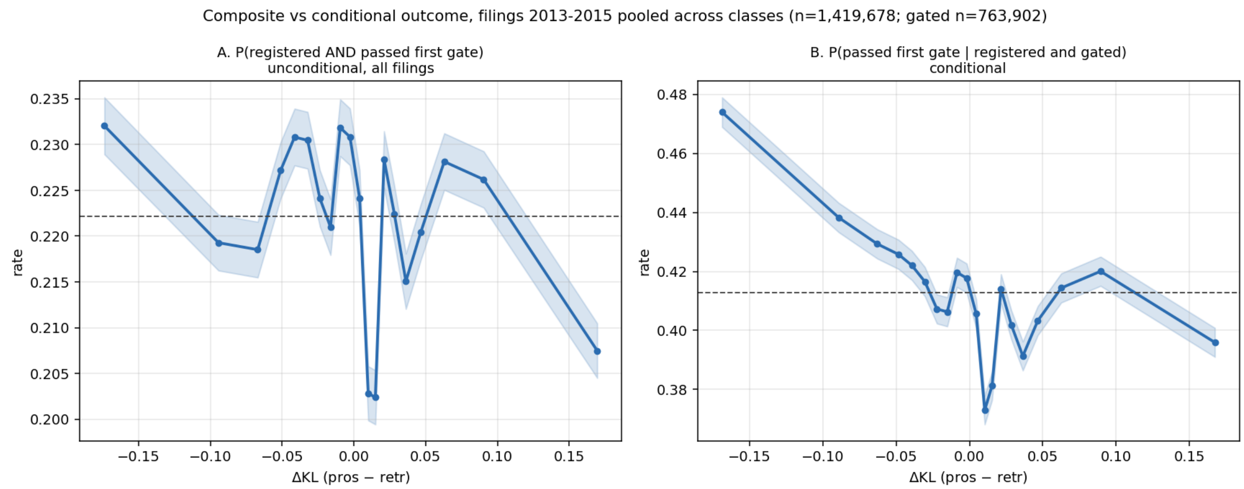

Figure 11: Filings 2013–2015 pooled across classes, topic-scored on per-filing reference windows ($n = 1{,}419{,}678$; gated subset $n = 763{,}902$). *A:* the unconditional composite P(registered and passed first gate) over all filings. *B:* P(passed | registered and gated) on the same filings. The unconditional view is nearly flat in lead (22.4% to 22.1% across quintiles); conditioning on grant is what exposes the gradient.

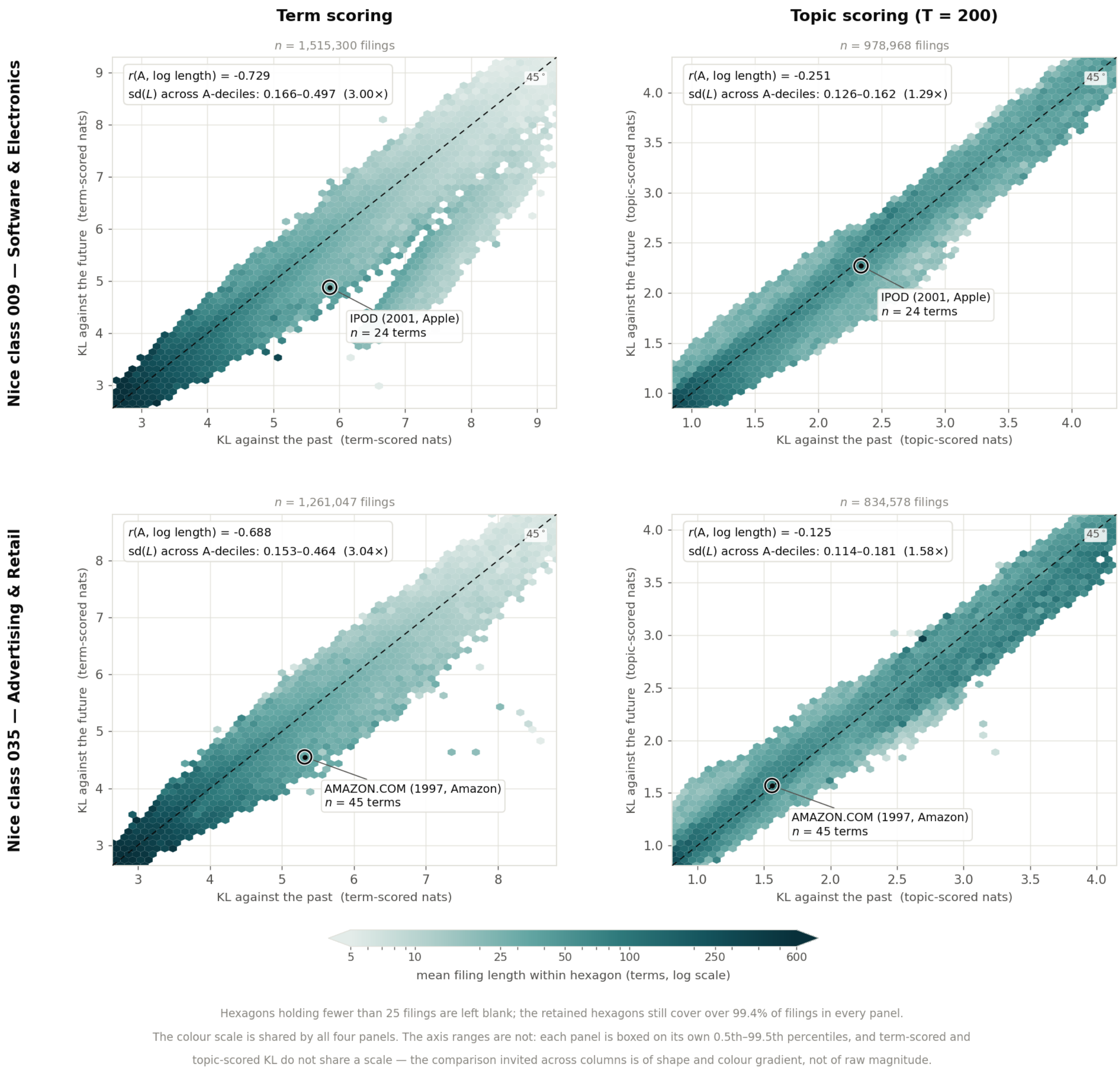

Figure 12: Mean filing length across the two axes, by representation and class. Hexagons holding fewer than 25 filings are blank; the retained hexagons cover over 99.4% of filings in every panel. The colour scale is shared across panels; the axis ranges are not, since term-scored and topic-scored KL do not share a scale, so the comparison invited across columns is of shape and colour gradient rather than of magnitude. Topic scoring returns a finite value for roughly 65% of filings, so the columns are not the same sample; the correlations quoted in Appendix A are re-estimated on the common subset and are substantially unchanged.